Faithful Self-Modeling Is Complex Quantum Mechanics

Bryan Ehrlich*

Abstract

Faithful self-modeling *is* complex quantum mechanics. We formalize self-modeling as four conditions on a finite-capacity system (Definition 2.6)—none of which mention complex numbers, Hilbert spaces, or Born probabilities—and prove that every self-modeling system is isomorphic to $M_n(\mathbb{C})^{\mathrm{sa}}$ with the Lüders product. The complex field, C^*-involution, and full algebraic structure of quantum theory are consequences of the definition, not inputs. The proof builds a sequential product from the system's own internal structure, verifies van de Wetering's axioms to obtain Jordan algebra structure, then shows that only complex systems admit the body–model composite the definition requires—excluding every non-complex alternative. Where other reconstruction programs require 3–6 independent axioms and typically assume the complex field or local tomography as input, here the complex field is derived and local tomography follows from the internal-accessibility requirement built into the definition. The downstream derivation chain (EJA classification through C^*-promotion) is machine-verified in Lean 4 with Mathlib (0 `sorry`); three of its four axioms correspond to published theorems of van de Wetering and Alfsen–Shultz, and one bridges the novel construction of Sections 3–4 to the formalized framework.

1 Introduction

Faithful self-modeling and complex quantum mechanics are the same thing. We formalize self-modeling as four conditions on a finite-dimensional system—spectral state space, faithful internal tracking, minimal body–model composite, and irreducibility—none of which mention complex numbers, Hilbert spaces, or C^*-algebras. We then prove that every system satisfying these

*Independent researcher. Electronic mail: bryan@ehrlich.dev

conditions is isomorphic to $M_n(\mathbb{C})^{\mathrm{sa}}$, the self-adjoint $n \times n$ complex matrices equipped with the Lüders sequential product $a \cdot b = \sqrt{a}\, b \sqrt{a}$ and the conjugate-transpose involution $X^* = X^\dagger$. The complex field, Born-rule structure, and C^*-involution all emerge as consequences of the self-modeling definition.

This result contributes to the quantum reconstruction program, which seeks to derive the mathematical framework of quantum theory from physically transparent principles. Beginning with Hardy's five-axiom reconstruction [Har01], the program has produced a succession of results that characterize quantum theory within the landscape of generalized probabilistic theories (GPTs): Chiribella, D'Ariano, and Perinotti [CDP11] used six axioms including Purification; Dakić and Brukner [DB11] used four axioms including local tomography; Masanes and Müller [MM11, MMAPG13] distilled the framework to three postulates assuming tomographic locality; and Barnum, Müller, and Ududec [BMU14] used four postulates including energy observability. A comprehensive review of the landscape is given by Müller [M21]. A common feature of these programs is the use of multiple independent operational axioms, and in most cases the complex number field—or a property that selects it, such as local tomography—is assumed rather than derived.

Our main result can be stated informally as follows.

Theorem 1.1 (Main result, informal)**.** *Every self-modeling system (Definition 2.6) is isomorphic to $M_n(\mathbb{C})^{\mathrm{sa}}$ for some $n \geq 2$, equipped with the Lüders sequential product and conjugate-transpose involution.*

The converse is immediate: $M_n(\mathbb{C})^{\mathrm{sa}}$ admits a faithful self-model (take $\varphi = \mathrm{id}$, with the standard tensor product as composite). The result is therefore a *characterization*: among finite-dimensional spectral order unit spaces, self-modeling systems are exactly the complex matrix algebras. The content of the theorem is the forward direction—that four conditions about self-modeling, none of which mention complex numbers or Hilbert spaces, turn out to characterize exactly the systems that have them.

The derivation proceeds through a chain of implications: self-modeling $\to$ sequential product construction $\to$ axiom verification $\to$ Euclidean Jordan algebra classification $\to$ local tomography $\to$ type exclusion $\to$ C^*-algebra promotion $\to$ involution exhibition. Figure 1 illustrates this chain, distinguishing the novel contributions of this paper from the published theorems invoked.

Table 1 compares the premise count and complex-selection mechanism of this work against the major reconstruction programs.

Table 1: Comparison of quantum reconstruction programs. $\mathbb{C}$: complex field; Born: Born-rule structure; C^*: full C^*-algebraic structure including involution. A = assumed, D = derived. Framework: GPT = generalized probabilistic theory, OPT = operational-probabilistic theory, OUS = order unit space with Alfsen–Shultz compressions. Our Definition 2.6 packages four conditions (see text); the richer OUS framework is part of the input.

Program	Input	$\mathbb{C}$	Born	C^*	Framework
Hardy [Har01]	5 axioms	A	D	A	GPT
CDP [CDP11]	6 axioms	D	D	A	OPT
Masanes–Müller [MMAPG13]	3 postulates	A	D	A	GPT
BMU [BMU14]	4 postulates	D	D	A	GPT
This work	**Def. 2.6 (4 conds.)**	**D**	**D**	**D**	**OUS**

The paper is organized as follows. Section 2 introduces order unit spaces, compressions, Peirce decompositions, sequential products, and the self-modeling definition. Section 3 presents the main novel construction: the self-modeling sequential product, the naive extension and its failure, the corrected product via Peirce feedback, and the determination of the mixing function. Section 4 verifies that the corrected product satisfies axioms S1–S7 of van de Wetering [vdW19b]. Section 5 constructs the body–model composite, proves local tomography, and addresses the entangled sector. Section 6 excludes all non-complex Euclidean Jordan algebra types and promotes the algebra to a C^*-algebra via the theorems of van de Wetering, Barnum–Wilce, and Hanche-Olsen. Section 7 discusses the scope and limitations of the result.

We close this introduction with an honest statement of scope. Definition 2.6 formalizes "self-modeling system" as four conditions, each unpacking one word of the phrase: *system* (finite-dimensional, spectral), *modeling* (faithful tracking), *self-* (internal, minimal composite), and *the whole* (simple, irreducible). Remove any condition and the concept of self-modeling breaks; add any condition and you assume more than self-modeling requires. Section 7.3 develops this argument in detail. The definition characterizes *composable* self-modelers (subsystems that can participate in composites); the non-composable case (the "self-modeling basin") is treated in a companion paper [Ehr26].

2 Preliminaries

2.1 Order Unit Spaces

We work within the framework of order unit spaces, following Alfsen–Shultz [AS03].

Definition 2.1 (Order unit space)**.** An *order unit space* $(V, V^+, \mathbb{1})$ is a real ordered vector space with a proper cone V^+ (closed, convex, pointed) generating V, equipped with a distinguished Archimedean order unit $\mathbb{1}$: for every $a \in V$ there exists $r \geq 0$ with $-r\mathbb{1} \leq a \leq r\mathbb{1}$. The *order unit norm* is $\|a\| = \inf\{r \geq 0 : -r\mathbb{1} \leq a \leq r\mathbb{1}\}$.

The *effect space* of V is $[0,1]_V = \{a \in V : 0 \leq a \leq \mathbb{1}\}$. An effect $p \in [0,1]_V$ is *sharp* (a *projective unit*) if p and $\mathbb{1} - p$ are both extremal in $[0,1]_V$, or equivalently if p admits no non-trivial refinement. We write $\mathrm{Proj}(V)$ for the set of projective units.

Every effect in a spectral order unit space admits a *spectral decomposition* $a = \sum_i \lambda_i \, p_i$, where the p_i are mutually orthogonal projective units with $\sum_i p_i = \mathbb{1}$ and the λ_i are distinct elements of $[0,1]$. This decomposition exists as a property of the OUS lattice structure [AS03], prior to and independent of any sequential product.

2.2 Compressions and Peirce Decomposition

For each projective unit $p \in \mathrm{Proj}(V)$, the Alfsen–Shultz theory provides a *compression* $C_p : V \to V$, a positive linear projection satisfying $C_p(\mathbb{1}) = p$. The compression C_p projects onto the face generated by p, and it plays the role of the Lüders operation $b \mapsto p\,b\,p$ in the general OUS setting.

Definition 2.2 (Peirce decomposition)**.** For a projective unit $p \in \mathrm{Proj}(V)$, the *Peirce decomposition* of V relative to p is the direct-sum decomposition

$$V = V_2(p) \oplus V_1(p) \oplus V_0(p), \tag{1}$$

where $V_2(p) = \mathrm{range}(C_p)$ is the face of p, $V_0(p) = \mathrm{range}(C_{p^\perp})$ is the face of $p^\perp = \mathbb{1} - p$, and $V_1(p) = \ker(C_p) \cap \ker(C_{p^\perp})$ is the *Peirce* 1*-space*.

A key identity is:

$$C_p + C_{p^\perp} = \mathrm{pinch}_p \quad \text{(the pinching map, } \textit{not} \text{ the identity).} \tag{2}$$

The pinching map annihilates $V_1(p)$. In the commutative (classical) case $V_1(p) = \{0\}$, so the pinching map is the identity; in the non-commutative

case $V_1(p) \neq \{0\}$ and the identity fails. This distinction drives the entire construction in Section 3.

More generally, for a resolution of unity $\{p_1, \ldots, p_n\}$ (mutually orthogonal projective units with $\sum_i p_i = \mathbb{1}$), the *Peirce* (i, j) *projection* P_{ij} extracts the component of b in the (i, j) Peirce 1-space $V_1(p_i, p_j)$:

$$P_{ij}(b) \in V_1(p_i, p_j), \qquad b = \sum_i C_{p_i}(b) + \sum_{i<j} P_{ij}(b). \tag{3}$$

2.3 Sequential Products

Following Gudder and Greechie [GG02] and van de Wetering [vdW19b], we define a sequential product on an order unit space.

Definition 2.3 (Sequential product space, [vdW19b, Def. 2])**.** Let $(V, \leq, \mathbb{1})$ be an order unit space equipped with a binary operation $(\cdot) : [0, 1]_V \times [0, 1]_V \to [0, 1]_V$. Write $a \mid b$ ("a and b are compatible") when $a \cdot b = b \cdot a$. Then $(V, \cdot)$ is a *sequential product space* if the following axioms hold:

(S1) **Additivity:** $a \cdot (b + c) = a \cdot b + a \cdot c$ whenever $b + c \leq \mathbb{1}$.

(S2) **Continuity:** The map $a \mapsto a \cdot b$ is continuous in the order unit norm.

(S3) **Unitality:** $\mathbb{1} \cdot a = a$ for all $a \in [0, 1]_V$.

(S4) **Symmetry of orthogonality:** If $a \cdot b = 0$ then $b \cdot a = 0$.

(S5) **Associativity of compatible effects:** If $a \mid b$ then $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.

(S6) **Additivity of compatibility:** If $a \mid b$ then $a \mid (\mathbb{1} - b)$; if also $a \mid c$ then $a \mid (b + c)$ whenever $b + c \leq \mathbb{1}$.

(S7) **Multiplicativity of compatibility:** If $a \mid b$ and $a \mid c$ then $a \mid b \cdot c$.

Two central results of van de Wetering are used as black boxes throughout this paper.

Theorem 2.4 ([vdW19b, Thm. 1])**.** *A finite-dimensional sequential product space is order-isomorphic to a Euclidean Jordan algebra.*

Theorem 2.5 ([vdW19b, Thm. 3])**.** *If a sequential product space has a locally tomographic composite that also carries a sequential product, then the sequential product space embeds into a C^*-algebra.*

The Jordan–von Neumann–Wigner classification [JvNW34] of simple finite-dimensional Euclidean Jordan algebras yields five types: $M_n(\mathbb{C})^{\mathrm{sa}}$ over $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$; the spin factors V_n; and the exceptional Albert algebra $M_3(\mathbb{O})^{\mathrm{sa}}$. Our task is to show that self-modeling is complex quantum mechanics.

2.4 Self-Modeling Systems

We now introduce the central definition.

Consider a physical system B (the "body") described by a finite-dimensional spectral OUS $(V_B, V_B^+, \nVdash_B)$. A *self-model* of B is a second system M (the "model"), described by an OUS $(V_M, V_M^+, \nVdash_M)$, together with a positive unital map $\varphi : V_B \to V_M$ (the *tracking map*) that encodes the operational cycle:

(i) *Prepare:* the body B is in some state;

(ii) *Update:* testing a sharp effect p on B triggers M to undergo the compression $C_{\varphi(p)}$, updating the model;

(iii) *Measure:* a second effect b is tested on B.

The composite outcome of this cycle—"first p, then b"—defines a binary operation on the effect space of B.

Definition 2.6 (Self-modeling system)**.** A *self-modeling system* is a tuple (V, φ, V_{BM}) satisfying:

(i) V is a nontrivial finite-dimensional spectral order unit space (i.e., V has at least two orthogonal nontrivial projective units);

(ii) $\varphi : V_B \to V_M$ is an order isomorphism (faithful tracking);

(iii) V_{BM} is the minimal composite OUS carrying product states, product effects, non-signaling constraints, and product-form sequential product;

(iv) V has no nontrivial direct-sum decomposition: there is no splitting $V = V_1 \oplus V_2$ into nonzero order-unit subspaces.

Each clause unpacks one word of the concept:

- *System* (i): has states, measurements, distinguishable outcomes. Finite because a self-modeler in a finite region has finite information capacity [Bek81]. Nontriviality excludes the degenerate case $V \cong \mathbb{R}$ (no nontrivial projective units, vacuous Peirce structure): a system with nothing to distinguish has nothing to model. All non-complex types with nontrivial projective units are excluded by the local tomography argument (Theorem 5.8), not by the definition.
- *Modeling* (ii): the internal model is accurate—an order isomorphism preserves all operationally accessible structure.

- *Self-* (iii): the model is internal. The only measurements available to the self-modeler are product measurements on the body and model. If the body–model composite carried states that product measurements could not distinguish, those states would be opaque to the self-modeler—aspects of its own composite that it cannot probe or model. Minimality says: the composite has no such hidden structure. The product-form sequential product (Definition 5.2) encodes non-signaling: testing the body and updating the model are independent operations on separate subsystems.
- *The whole* (iv): a direct-sum algebra decomposes into sectors invisible to each other; a system in one sector cannot model another. Simplicity is the algebraic expression of "whole models whole."

Remove any clause and the concept of self-modeling breaks. Add any clause and you have assumed more than self-modeling requires.

Remark 2.7. Approximate or partial self-models (where φ is merely positive or injective but not surjective) might yield weaker algebraic structures; we do not pursue this here. Definition 2.6 characterizes *composable* self-modelers—subsystems that can participate in composites. The non-composable case ($h_3(\mathbb{O})$, which admits no composite at all) is treated in a companion paper [Ehr26].

3 The Self-Modeling Sequential Product

3.1 From Self-Modeling to Sequential Product

The self-modeling cycle of Section 2.4 produces a natural binary operation on the effects of B. For sharp effects $p, q \in \mathrm{Proj}(V_B)$, the composite outcome "first test p, update the model, then test q" is encoded by the compression product:

$$p \cdot q = C_p(q), \tag{4}$$

where C_p is the Alfsen–Shultz compression for the face generated by p in V_B.

Several properties of this product follow immediately from the compression axioms. It maps effects to effects (positivity of C_p). It satisfies $\nVdash \cdot q = q$ (since $C_{\nVdash}$ is the identity). On a simplex (classical state space), it reduces to the pointwise product $p \cdot q = (p_1 q_1, \ldots, p_n q_n)$, matching the unique sequential product on simplices [GG02].

Remark 3.1. The product (4) is defined entirely on V_B using B's own compression structure. The tracking map φ provides the physical justification—

testing p on B causes the model M to undergo $C_{\varphi(p)}$, which feeds back to the context for testing q—but the algebraic product is intrinsic to B.

3.2 The Naive Extension and Its Failure

To extend the product from sharp to general effects, one attempts a spectral extension. Given $a = \sum_i \lambda_i \, p_i$ (spectral decomposition in the OUS), define:

$$a \cdot b \stackrel{?}{=} \sum_i \lambda_i \, C_{p_i}(b). \tag{5}$$

This is the natural extension: the effect a acts on b as a weighted sum of compressions, one for each spectral projector.

However, this extension *fails* axiom (S3) (unitality) on non-commutative order unit spaces. To see why, set $a = \mathbb{1}$ with spectral decomposition $\mathbb{1} = \sum_i p_i$ (all eigenvalues equal to 1). Then (5) gives

$$\mathbb{1} \cdot b = \sum_i C_{p_i}(b) = \mathrm{pinch}_{p_1,\ldots,p_n}(b) \neq b \quad \text{in general.} \tag{6}$$

The sum of compressions is the *pinching map*, not the identity. By the Peirce decomposition (3), the pinching map annihilates all Peirce 1-space components: $P_{ij}(b)$ is discarded for every $i < j$. In the commutative case $V_1 = \{0\}$, the pinching map *is* the identity and the extension succeeds—this is why the classical limit works. In the non-commutative case, the Peirce 1-space is nontrivial, and the naive extension destroys off-diagonal information.

This failure is not a bug; it reveals a structural gap. The compression sum captures the Peirce 2-space (diagonal) contributions correctly but provides no prescription for the Peirce 1-space (off-diagonal) terms. Closing this gap requires new input.

3.3 The Corrected Product via Peirce Feedback

The missing Peirce 1-space contribution comes from the self-modeling feedback loop. We now show that the form of the corrected product is *forced* by the axioms, not chosen. Any binary operation on effects satisfying (S1) (additivity in the second argument), (S2) (continuity in the first), and the sharp constraint $p \cdot b = C_p(b)$ must respect the Peirce decomposition: the map $a \cdot \cdot$ is a linear endomorphism (by (S1) and finite-dimensionality) whose range on each Peirce subspace is constrained to that subspace (by the compression algebra structure). The diagonal coefficients are pinned to the eigenvalues by

eigenvalue coalescence and (S2). Thus the general product for $a = \sum_i \lambda_i p_i$ must take the form:

$$a \cdot b = \sum_i \lambda_i C_{p_i}(b) + \sum_{i<j} f(\lambda_i, \lambda_j) P_{ij}(b), \tag{7}$$

where the first sum captures diagonal (Peirce 2-space) contributions via compressions, and the second sum restores the off-diagonal (Peirce 1-space) contributions through a *mixing function* $f : [0,1] \times [0,1] \to \mathbb{R}$ to be determined.

Why this form is forced. By (S1), $a \cdot\cdot$ is a linear endomorphism of V. The Peirce decomposition with respect to the spectral projectors of a decomposes V into subspaces $V_2(p_i)$ and $V_1(p_i, p_j)$. Compressions project onto these subspaces (Alfsen–Shultz [AS03]), so linearity gives a block decomposition: $a\cdot\cdot$ maps each Peirce subspace to itself. On $V_2(p_i)$, the sharp constraint $p_i \cdot b = C_{p_i}(b)$ and coalescence ($\lambda_i \to \lambda_j$ via (S2)) fix the action to $\lambda_i C_{p_i}(\cdot)$. On $V_1(p_i, p_j)$, the action is a linear endomorphism $E_{ij}(\lambda_i, \lambda_j)$ of $V_1(p_i, p_j)$, depending continuously on the eigenvalues by (S2). In Proposition 3.3, we prove that compatible associativity ((S5)), eigenvalue symmetry, and coalescence force E_{ij} to be a scalar multiple of the identity on each $V_1(p_i, p_j)$: that is, $E_{ij}(\lambda_i, \lambda_j) = f(\lambda_i, \lambda_j) \cdot P_{ij}$ for a single mixing function f, simultaneously determining $f = \sqrt{\lambda_i \lambda_j}$. Thus (7) is the unique form compatible with (S1)–(S5) and the sharp constraint.

The mixing function is constrained by the requirement that the product map effects to effects: $0 \leq a \cdot b \leq \mathbb{1}$ for all effects a, b. This yields a *positivity bound.*

Proposition 3.2 (Positivity bound)**.** *For the product* (7) *to map* $[0,1]_V \times [0,1]_V$ *into* $[0,1]_V$*, the mixing function must satisfy*

$$|f(\lambda_i, \lambda_j)| \leq \sqrt{\lambda_i \lambda_j} \tag{8}$$

for all $\lambda_i, \lambda_j \in [0,1]$.

Proof. Every pair of orthogonal atoms p_i, p_j in a spectral OUS generates a two-level face isomorphic to a spin factor, on which the Schur complement criterion gives $|f(\lambda_i, \lambda_j)| \leq \sqrt{\lambda_i \lambda_j}$. Face effects embed into global effects, so the bound lifts to the full space. □

For the saturating choice $f = \sqrt{\lambda_i \lambda_j}$ (Proposition 3.3), the product maps effects to effects globally: the coefficient matrix $[M_{ij}]$ with $M_{ij} = \sqrt{\lambda_i \lambda_j}$ factors as vv^T with $v_i = \sqrt{\lambda_i}$, hence is positive semidefinite. Together

with positivity of compressions, this gives $a \cdot b \geq 0$ when $a, b \geq 0$; the upper bound $a \cdot b \leq \mathbb{1}$ follows from (S1) (additivity in b) and (S3) (unitality), both verified in Section 4.

3.4 Determining the Mixing Function

As shown in Section 3.3, the self-modeling construction provides a product whose action on each Peirce 1-space $V_1(p_i, p_j)$ is an undetermined linear endomorphism $E_{ij}(\lambda_i, \lambda_j)$. We now show that internal consistency forces E_{ij} to be scalar and uniquely determines the scalar.

Proposition 3.3 (Scalar action and maximal coherence)**.** *The action of $a \cdot \cdot$ on each Peirce 1-space $V_1(p_i, p_j)$ is scalar multiplication by $\sqrt{\lambda_i \lambda_j}$:*

$$E_{ij}(\lambda_i, \lambda_j) = \sqrt{\lambda_i \lambda_j} \cdot \mathrm{Id}_{V_1(p_i, p_j)}. \tag{9}$$

In particular, the mixing function is $f(\lambda_i, \lambda_j) = \sqrt{\lambda_i \lambda_j}$ and the product has the form (7)*.*

Proof. The argument treats (S5) as a *consistency constraint*: we seek the endomorphisms E_{ij} for which the resulting product satisfies (S5). (Once E_{ij} is determined, Section 4 verifies S1–S7 in full; the logic is not circular.)

Step 1: Multiplicative composition. For compatible $a = \alpha_1 p_1 + \alpha_2 p_2$ and $b = \beta_1 p_1 + \beta_2 p_2$ sharing an eigenbasis, imposing (S5)—i.e., requiring $a \cdot (b \cdot d) = (a \cdot b) \cdot d$ for all effects d—and comparing the action on $V_1(p_1, p_2)$ gives

$$E(\alpha_1, \alpha_2) \circ E(\beta_1, \beta_2) = E(\alpha_1 \beta_1,\ \alpha_2 \beta_2), \tag{10}$$

where $E = E_{12}$ and $\circ$ denotes composition of endomorphisms of $V_1(p_1, p_2)$.

Step 2: Factorization. Define $h(x) = E(x, 1)$ and $k(y) = E(1, y)$. Equation (10) gives $h(x_1)h(x_2) = h(x_1 x_2)$ and $k(y_1)k(y_2) = k(y_1 y_2)$, so h and k are continuous multiplicative maps $(0, 1] \to \mathrm{End}(V_1)$. Setting $\alpha_2 = \beta_1 = 1$ gives $E(x, y) = h(x)\, k(y)$, and h, k commute (both orderings give $E(x, y)$). Every continuous multiplicative homomorphism $(0, 1] \to \mathrm{GL}(W)$ on a finite-dimensional space W has the form $t \mapsto t^A = \exp(A \ln t)$ for some $A \in \mathrm{End}(W)$ (Aczél [Acz66], Ch. 4). Applied to h and k: $h(x) = x^A$ and $k(y) = y^B$ for matrices A, B with $[A, B] = 0$.

Step 3: Symmetry. The Peirce 1-space $V_1(p_1, p_2) = V_1(p_2, p_1)$ is the same subspace regardless of index order. The product sums over unordered pairs, so $E(x, y) = E(y, x)$ as endomorphisms. This gives $x^A\, y^B = y^A\, x^B$ for all $x, y > 0$, which forces $A = B$. Hence $E(x, y) = (xy)^A$.

Step 4: Coalescence. By (S2), as $\lambda_1 \to \lambda_2 = \lambda$, the product reduces to $(\lambda \Vdash) \cdot b = \lambda b$, giving

$$E(\lambda, \lambda) = \lambda \cdot \mathrm{Id} \qquad \text{for all } \lambda \in (0, 1]. \tag{11}$$

Substituting $E(\lambda, \lambda) = (\lambda^2)^A = \lambda^{2A}$ and comparing with $\lambda\,\mathrm{Id} = \lambda^{\mathrm{Id}}$ gives $2A = \mathrm{Id}$, so $A = \frac{1}{2}\,\mathrm{Id}$. Therefore $E(x, y) = (xy)^{\mathrm{Id}/2} = \sqrt{xy}\,\mathrm{Id}$.

The endomorphism is scalar, with $f(\lambda_i, \lambda_j) = \sqrt{\lambda_i\,\lambda_j}$. □

To clarify the logical structure: Section 3.3 shows that linearity, continuity, and the sharp constraint force the product into a block-diagonal form with an undetermined endomorphism on each Peirce 1-space. The present proposition shows that *requiring* (S5) (a necessary condition for any sequential product space) forces the endomorphism to be scalar and uniquely determines the scalar. Section 4 then verifies all of S1–S7 for the resulting product with $f = \sqrt{\lambda_i\,\lambda_j}$, closing the loop. The role of self-modeling is constructive: the tracking isomorphism φ provides the product on which the axioms act. Without φ, there is no product to constrain; once the product exists, the axioms determine f uniquely.

The corrected product with $f = \sqrt{\lambda_i\,\lambda_j}$ is:

$$\boxed{a \cdot b = \sum_i \lambda_i\, C_{p_i}(b) + \sum_{i<j} \sqrt{\lambda_i\,\lambda_j}\, P_{ij}(b).} \tag{12}$$

Remark 3.4. On $M_2(\mathbb{C})^{\mathrm{sa}}$, the corrected product (12) coincides with the Lüders product $a \cdot b = \sqrt{a}\,b\,\sqrt{a}$. This equivalence is a *consequence* of the construction, not a premise. The Lüders formula was never assumed; it emerges from the positivity bound and faithful self-modeling.

Remark 3.5 (Division of labor). The tracking map φ and the sequential-product axioms play complementary roles. Self-modeling provides the operational justification for a binary product on effects (the test–update–test cycle) and constrains its form to the ansatz (7). The axioms—specifically (S5)—then pin the mixing function uniquely. Neither ingredient suffices alone: without φ, there is no product to constrain; without (S5), the mixing function is underdetermined. Heuristically, $f = 0$ corresponds to a trivial (non-faithful) model and $f = \sqrt{\lambda_i\,\lambda_j}$ to a faithful one, but we do not prove a quantitative theorem linking the fidelity of φ to the value of f; the connection is that faithfulness of φ justifies seeking the product that saturates the positivity bound (8), and (S5) identifies that product uniquely.

The remaining clauses of Definition 2.6—minimality (iii) and simplicity (iv)—enter in Section 5.

3.5 Circularity Check

A natural concern is whether the scalar function $\sqrt{\lambda_i \lambda_j}$ in (12) imports C^*-algebraic structure through the back door. It does not. The square root in the mixing function operates on *real eigenvalues* $\lambda_i, \lambda_j \in [0,1]$: it is arithmetic, specifically $\sqrt{x} : [0,1] \to [0,1]$, the unique non-negative real number whose square is x. No operator square root, no trace, no inner product, no density matrix, and no Hilbert space structure is used in the construction. The only mathematical objects employed are:

- the order unit space $(V, V^+, \nVdash)$;
- the effect space $[0,1]_V$ and its projective units;
- Alfsen–Shultz compressions C_p for faces of projective units;
- the Peirce decomposition (3) derived from compressions;
- spectral decomposition of effects in the OUS;
- the scalar function $\sqrt{\cdot}$ on $[0,1] \subset \mathbb{R}$.

Complex numbers, Hilbert spaces, and C^*-algebraic operations do not appear until the conclusion (Section 6), where they emerge as consequences of the classification theorems.

4 Axiom Verification

We now verify that the self-modeling sequential product defined in Section 3 satisfies all seven axioms of van de Wetering's Definition 2 [vdW19b]. By Theorem 1 of [vdW19b], this establishes that the state space carries the structure of a Euclidean Jordan algebra.

Throughout this section, let V be a finite-dimensional spectral order unit space and let the sequential product be given by the corrected formula (12):

$$a \cdot b = \sum_i \lambda_i \, C_{p_i}(b) + \sum_{i<j} \sqrt{\lambda_i \lambda_j} \, P_{ij}(b), \tag{13}$$

where $a = \sum_i \lambda_i \, p_i$ is the spectral decomposition of a, C_{p_i} is the Alfsen–Shultz compression for the face generated by p_i, and P_{ij} is the Peirce (i,j) projection onto the 1-space $V_1(p_i, p_j)$ as defined in (3).

S1 (Additivity)

The map $b \mapsto a \cdot b$ is additive: $a \cdot (b + c) = a \cdot b + a \cdot c$ whenever $b + c \leq \mathbb{1}$.

Each compression C_{p_i} is a positive linear map on V (Alfsen–Shultz [AS03], Ch. 7), and the Peirce projection P_{ij} is linear (each P_{ij} is a linear combination of compressions by (3)). Since the coefficients λ_i and $\sqrt{\lambda_i \lambda_j}$ depend only on a, the map $b \mapsto a \cdot b$ is a fixed linear combination of linear maps applied to b, and therefore additive. □

S2 (Continuity)

The map $a \mapsto a \cdot b$ is continuous in the order unit norm.

The eigenvalues $\lambda_i(a)$ are continuous functions of a in finite dimensions. The eigenprojectors $p_i(a)$, however, are *discontinuous* at degeneracies: when two eigenvalues coalesce, the spectral projectors jump. The product formula nonetheless depends continuously on a because the mixing function smooths the discontinuity. At a degeneracy $\lambda_i = \lambda_j$, the off-diagonal coefficient $\sqrt{\lambda_i \lambda_j} = \lambda_i$ equals the diagonal coefficients, so the Peirce 1-space term $\sqrt{\lambda_i \lambda_j}\, P_{ij}(b)$ merges continuously with the diagonal terms $\lambda_i\, C_{p_i}(b)$. Formally, the product $a \cdot b$ equals the spectral functional calculus $g_b(a) := \sum_i \lambda_i\, C_{p_i}(b) + \sum_{i<j} \sqrt{\lambda_i \lambda_j}\, P_{ij}(b)$, which is a continuous function of the eigenvalues alone (Alfsen–Shultz [AS03], Ch. 9, Thm. 9.37). Since the eigenvalues depend continuously on a, the composition is continuous. □

S3 (Unitality)

$\mathbb{1} \cdot a = a$ for all effects a.

The order unit $\mathbb{1}$ has the trivial spectral decomposition $\mathbb{1} = 1 \cdot \mathbb{1}$ (a single eigenvalue with the maximal projective unit). The Peirce 1-space sum is empty, and the compression $C_{\mathbb{1}}$ for the full face is the identity map on V (Alfsen–Shultz [AS03], Def. 7.1). Therefore $\mathbb{1} \cdot a = 1 \cdot C_{\mathbb{1}}(a) = a$.

This is the axiom whose verification critically depends on the Peirce 1-space correction: the naive product $\sum_i \lambda_i\, C_{p_i}(b)$ (without the $\sqrt{\lambda_i \lambda_j}\, P_{ij}(b)$ terms) gives $\mathbb{1} \cdot a = \mathrm{pinch}(a) \neq a$ for effects with off-diagonal components, because the sum of compressions over a resolution of unity is the pinching map, not the identity. The self-modeling feedback that introduces the Peirce 1-space terms is what restores unitality. □

S4 (Orthogonality Symmetry)

If $a \cdot b = 0$ then $b \cdot a = 0$.

This is the key axiom. The remaining axioms (S1–S3, S5–S7) follow from linearity and spectral calculus with comparatively short arguments. S4 requires a substantive proof that exploits the facial structure of the order unit space. We give the argument in full; see Appendix A.1 for additional details.

Theorem 4.1. *Let $(V, \leq, \mathbb{1})$ be a finite-dimensional spectral order unit space equipped with the sequential product* (13)*. If $a \cdot b = 0$ for effects $a, b \in [0, \mathbb{1}]_V$, then $b \cdot a = 0$.*

Proof sketch. Write $a = \sum_{i=1}^{n} \lambda_i \, p_i$ (spectral decomposition). Two cases arise.

Case A (full-rank). Suppose all $\lambda_i > 0$. The condition $a \cdot b = 0$ and the direct sum structure of the Peirce decomposition

$$V = \bigoplus_i V_2(p_i) \;\oplus\; \bigoplus_{i<j} V_1(p_i, p_j)$$

(Alfsen–Shultz [AS03], Theorem 9.37) force each component to vanish individually: $\lambda_i \, C_{p_i}(b) = 0$ and $\sqrt{\lambda_i \lambda_j} \, P_{ij}(b) = 0$. Since $\lambda_i > 0$ for all i, the completeness of the Peirce decomposition gives $b = 0$, so $b \cdot a = 0$ trivially.

Case B (rank-deficient). Suppose $\lambda_1, \ldots, \lambda_m > 0$ and $\lambda_{m+1} = \cdots = \lambda_n = 0$ for some $1 \leq m < n$. As before, the Peirce direct sum forces $C_{p_i}(b) = 0$ for each $i \leq m$. The crucial step invokes the Alfsen–Shultz facial absorption theorem (Proposition 7.43 of [AS03]):

If $C_p(b) = 0$ and $b \geq 0$, then $b \in \mathrm{face}(p^\perp)$.

Applying this to each p_i ($i \leq m$) places b in the complementary face $\mathrm{face}\big(\sum_{k=m+1}^{n} p_k\big)$. By the facial structure of spectral order unit spaces, the Peirce 1-space components $V_1(p_i, p_j)$ connecting a face to its complement are excluded: an effect in $\mathrm{face}(p^\perp)$ has zero component in every $V_1(p_i, p_j)$ for $i \leq m$, $j > m$. Therefore b is supported entirely on the zero-eigenvalue block of a.

For the reverse product $b \cdot a$, write $b = \sum_j \mu_j \, q_j$ with each q_j ($\mu_j > 0$) lying in $\mathrm{face}\big(\sum_{k>m} p_k\big)$. Since a is supported on the complementary face $\mathrm{face}\Big(\sum_{i \leq m} p_i\Big)$, the facial orthogonality theorem gives $C_{q_j}(a) = 0$ for all

j with $\mu_j > 0$. The Peirce 1-space terms $Q_{jk}(a)$ either vanish by facial structure (when both $\mu_j, \mu_k > 0$) or carry zero weight $\sqrt{\mu_j \mu_k} = 0$. Hence $b \cdot a = 0$. □

Remark 4.2 (Independence of φ). Axiom S4 is φ-independent: it holds for *any* mixing function f with $f(0, x) = 0$, not just for the faithful choice $f = \sqrt{\lambda_i \lambda_j}$. The proof depends only on the facial geometry of the order unit space and the vanishing of f at zero eigenvalues, not on the specific form of f for positive arguments. Consequently, S4 is a structural property of spectral order unit spaces, not an artifact of the self-modeling construction.

S5 (Compatible Associativity)

> *If a and b are compatible (i.e., $a \cdot b = b \cdot a$), then $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all effects c.*

Compatible effects share a joint spectral decomposition. For sharp compatible effects p, q, the compressions commute: $C_p \circ C_q = C_q \circ C_p$ (Alfsen–Shultz [AS03], Prop. 7.49), and their composition satisfies $C_p \circ C_q = C_{p \wedge q}$ (Prop. 7.50).

For general compatible effects $a \mid b$, the Peirce vanishing lemma (Lemma 4.3) gives $P_1^{(a)}(b) = 0$, so b is block-diagonal in a's spectral decomposition. Let $a = \sum_i \alpha_i \, p_i$ and $b = \sum_j \beta_j \, q_j$ with the q_j refinements of the p_i (both resolutions commute). Then for any effect c:

$$a \cdot (b \cdot c) = \sum_{j,k} \alpha_{i(j)} \, \beta_j \, \sqrt{\beta_j \beta_k} \, C_{q_j}(P_{jk}(c)),$$

where $i(j)$ is the index with $q_j \leq p_{i(j)}$. Because q_j refines p_i, the coefficient factors as $\alpha_{i(j)} \sqrt{\beta_j \beta_k}$. The same expansion for $(a \cdot b) \cdot c$ yields the product $a \cdot b = \sum_j \alpha_{i(j)} \beta_j \, q_j$ (block-diagonal, since cross-terms vanish), and the identity $\sqrt{\alpha\beta \cdot \gamma\delta} = \sqrt{\alpha\gamma} \cdot \sqrt{\beta\delta}$ for non-negative reals gives term-by-term equality. □

S6 (Additivity of Compatibility)

> *If $a \mid b$ then $a \mid (\nVdash - b)$; if also $a \mid c$ then $a \mid (b + c)$ whenever $b + c \leq \nVdash$.*

Both parts of S6 rest on two preliminary observations that reduce compatibility to a Peirce vanishing condition.

Lemma 4.3 (Peirce vanishing)**.** *Let $a = \sum_i \lambda_i\, p_i$ and $b = \sum_j \mu_j\, q_j$ be effects in a finite-dimensional spectral OUS. If $a \mid b$, then b has zero Peirce 1-space component in a's spectral decomposition:*

$$P_1^{(a)}(b) \;=\; b - \sum_i C_{p_i}(b) \;=\; 0, \tag{14}$$

and symmetrically $P_1^{(b)}(a) = 0$.

Proof. By Alfsen–Shultz [AS03] Prop. 7.49 and Niestegge [Nie08] Lemma 3.3, $a \mid b$ implies that the spectral projectors are pairwise compatible: $C_{p_i} \circ C_{q_j} = C_{q_j} \circ C_{p_i}$ for all i, j. By Prop. 7.50 of [AS03], $C_{p_i} \circ C_{q_j} = C_{p_i \wedge q_j}$. Summing over j:

$$\sum_j C_{q_j}(p_i) = \sum_j (p_i \wedge q_j) = p_i,$$

where the last equality uses the refinement property: since $\{q_j\}$ resolves $\mathbb{1}$ and each q_j is compatible with p_i, the meets $\{p_i \wedge q_j\}_j$ resolve p_i. Therefore $\sum_j C_{q_j}(a) = \sum_i \lambda_i \sum_j (p_i \wedge q_j) = a$, giving $P_1^{(b)}(a) = a - \sum_j C_{q_j}(a) = 0$. Interchanging a and b gives $P_1^{(a)}(b) = 0$. □

Proposition 4.4 (Block-diagonal compatibility)**.** *If $P_1^{(a)}(e) = 0$ for an effect e and $a = \sum_i \lambda_i\, p_i$, then $a \mid e$.*

Proof. Since $e = \sum_i C_{p_i}(e)$ with each $C_{p_i}(e) \in \mathrm{face}(p_i)$, the spectral decomposition of e is the union of spectral decompositions within each face: write $C_{p_i}(e) = \sum_\ell \delta_{i\ell}\, s_{i\ell}$ with $s_{i\ell} \le p_i$. Each $s_{i\ell}$ satisfies $C_{p_{i'}} \circ C_{s_{i\ell}} = C_{s_{i\ell}} \circ C_{p_{i'}}$ for all i' (they commute trivially: $s_{i\ell} \le p_i$ gives $C_{s_{i\ell}} \circ C_{p_i} = C_{s_{i\ell}}$, while $s_{i\ell} \perp p_{i'}$ for $i' \neq i$ gives $C_{s_{i\ell}} \circ C_{p_{i'}} = 0$, both symmetric under composition reversal). The corrected product formula then gives, using $C_{s_{i\ell}}(a) = \lambda_i\, s_{i\ell}$:

$$e \cdot a = \sum_{i,\ell} \delta_{i\ell}\, C_{s_{i\ell}}(a) = \sum_{i,\ell} \delta_{i\ell}\, \lambda_i\, s_{i\ell} = \sum_i \lambda_i\, C_{p_i}(e) = a \cdot e,$$

where the Peirce 1-space terms vanish on both sides. □

Part (i): $a \mid b \implies a \mid (\mathbb{1} - b)$. By Lemma 4.3, $P_1^{(a)}(b) = 0$. Since $P_1^{(a)}(\mathbb{1}) = 0$ (the unit lies in the Peirce 2-space) and $P_1^{(a)}$ is linear:

$$P_1^{(a)}(\mathbb{1} - b) = P_1^{(a)}(\mathbb{1}) - P_1^{(a)}(b) = 0.$$

By Proposition 4.4, $a \mid (\mathbb{1} - b)$. □

Part (ii): $a \mid b$ and $a \mid c \implies a \mid (b+c)$ when $b+c \leq \mathbb{1}$. By Lemma 4.3, $P_1^{(a)}(b) = 0$ and $P_1^{(a)}(c) = 0$. By linearity:

$$P_1^{(a)}(b+c) = P_1^{(a)}(b) + P_1^{(a)}(c) = 0.$$

By Proposition 4.4, $a \mid (b+c)$. □

S7 (Multiplicativity of Compatibility)

If $a \mid b$ and $a \mid c$ then $a \mid b \cdot c$.

We show that $b \cdot c$ has zero Peirce 1-space component in a's decomposition, then invoke Proposition 4.4.

Write $a = \sum_i \lambda_i \, p_i$, $b = \sum_j \mu_j \, q_j$, $c = \sum_k \nu_k \, r_k$. By Lemma 4.3, $a \mid b$ gives $C_{p_i} \circ C_{q_j} = C_{q_j} \circ C_{p_i}$ for all i, j, and $a \mid c$ gives $C_{p_i} \circ C_{r_k} = C_{r_k} \circ C_{p_i}$ for all i, k.

Step 1: C_{p_i} commutes with every Peirce projection of b's resolution. The Peirce (j, j') projection for b's resolution is $Q_{jj'} = C_{q_j+q_{j'}} - C_{q_j} - C_{q_{j'}}$ (a linear combination of compressions). Since p_i is compatible with both q_j and $q_{j'}$, Prop. 7.50 of [AS03] gives $C_{p_i} \circ C_{q_j+q_{j'}} = C_{p_i \wedge (q_j+q_{j'})} = C_{(q_j+q_{j'}) \wedge p_i} = C_{q_j+q_{j'}} \circ C_{p_i}$. Therefore C_{p_i} commutes with $Q_{jj'} = C_{q_j+q_{j'}} - C_{q_j} - C_{q_{j'}}$.

Step 2: Compute $\sum_i C_{p_i}(b \cdot c)$. Expanding $b \cdot c$ via (13) and using linearity:

$$\sum_i C_{p_i}\big(b \cdot c\big) = \sum_j \mu_j \, C_{q_j}\Big(\sum_i C_{p_i}(c)\Big) + \sum_{j<j'} \sqrt{\mu_j \mu_{j'}} \, Q_{jj'}\Big(\sum_i C_{p_i}(c)\Big),$$

where we used Step 1 to pull $\sum_i C_{p_i}$ through the maps C_{q_j} and $Q_{jj'}$. By Lemma 4.3, $a \mid c$ gives $\sum_i C_{p_i}(c) = c$. Therefore

$$\sum_i C_{p_i}\big(b \cdot c\big) = \sum_j \mu_j \, C_{q_j}(c) + \sum_{j<j'} \sqrt{\mu_j \mu_{j'}} \, Q_{jj'}(c) = b \cdot c.$$

Conclusion. $P_1^{(a)}\big(b \cdot c\big) = b \cdot c - \sum_i C_{p_i}\big(b \cdot c\big) = 0$. By Proposition 4.4, $a \mid b \cdot c$. □

With S1–S7 verified, we invoke the central classification result:

Theorem 4.5 (EJA classification; [vdW19b, Theorem 1])**.** *Let $(V, \leq, \mathbb{1})$ be a finite-dimensional order unit space admitting a sequential product satisfying S1–S7. Then V is order-isomorphic to a Euclidean Jordan algebra.*

By the Jordan–von Neumann–Wigner classification [JvNW34], the simple Euclidean Jordan algebras are: $M_n(\mathbb{R})^{\mathrm{sa}}$, $M_n(\mathbb{C})^{\mathrm{sa}}$, $M_n(\mathbb{H})^{\mathrm{sa}}$, the spin factors V_n, and the exceptional Albert algebra $M_3(\mathbb{O})^{\mathrm{sa}}$.

Remark 4.6 (Non-associativity). The sequential product $a{\cdot}b$ is non-associative for general (non-compatible) effects: there exist triples (a, b, c) with $(a{\cdot}b){\cdot}c \neq a \cdot (b \cdot c)$. This is consistent with S5, which requires associativity only for compatible pairs. By the theorem of Westerbaan, Westerbaan, and van de Wetering [WWvdW20], associativity would force commutativity, which would force the state space to be a simplex (classical). Non-associativity is thus a necessary feature of any non-classical sequential product.

5 Composite Systems and Local Tomography

5.1 The Body–Model Composite

Having established that the state space V of the body B (and isomorphically the model M) carries EJA structure, we now formalize the composite system V_{BM} using only generalized probabilistic theory (GPT) primitives [Pla23, BUvdW23].

Definition 5.1 (Composite OUS)**.** Let V_B and V_M be finite-dimensional order unit spaces, each equipped with a sequential product satisfying S1–S7. The *composite order unit space* V_{BM} is defined by:

(C1) **Vector space.** V_{BM} is a finite-dimensional real vector space containing $V_B \otimes V_M$ (the algebraic tensor product of real vector spaces) as a subspace.

(C2) **Order unit.** $\mathbb{1}_{BM} = \mathbb{1}_B \otimes \mathbb{1}_M$.

(C3) **Product states.** For every state ω_B on V_B and ω_M on V_M, the product state $\omega_B \otimes \omega_M$ belongs to $\mathcal{S}(V_{BM})$, defined by $(\omega_B \otimes \omega_M)(a \otimes b) = \omega_B(a)\,\omega_M(b)$.

(C4) **Non-signaling.** For every joint state $\omega \in \mathcal{S}(V_{BM})$, the marginals $\omega_B(a) \coloneqq \omega(a \otimes \mathbb{1}_M)$ and $\omega_M(b) \coloneqq \omega(\mathbb{1}_B \otimes b)$ are well-defined states, independent of what is measured on the other subsystem.

By Definition 2.6(iii), V_{BM} is the *minimal* order unit space satisfying (C1)–(C4) and the product-form sequential product (Definition 5.2 below).

This construction uses only real vector spaces, compact convex state spaces, affine functionals, and the non-signaling constraint — no Hilbert space tensor products, complex numbers, or density matrices.

5.2 Product-Form Sequential Product

Definition 5.2 (Product-form SP)**.** The sequential product on V_{BM} is defined on product effects by

$$(a \otimes b) \cdot (c \otimes d) = (a \cdot c) \otimes (b \cdot d), \tag{15}$$

where $\cdot\cdot\cdot$ on the right-hand side denotes the factor-level sequential product (13).

Proposition 5.3 (S1–S7 inheritance)**.** *If the factor sequential products on V_B and V_M each satisfy S1–S7, then the product-form sequential product on V_{BM} satisfies S1–S7.*

Proof sketch. Axioms S1 (additivity), S2 (continuity), and S3 (unitality) follow directly from the corresponding factor-level axioms and the bilinearity of the tensor product. Axioms S5–S7 (compatible associativity, additivity, multiplicativity) reduce to the factor-level axioms because compatibility of product effects $a \otimes b$ and $c \otimes d$ follows from factor-wise compatibility $a \cdot c = c \cdot a$ and $b \cdot d = d \cdot b$.

The inheritance of S4 (orthogonality symmetry) is more subtle. If $(a \otimes b) \cdot (c \otimes d) = 0$, then $(a \cdot c) \otimes (b \cdot d) = 0$. In an order unit space, states separate points (Alfsen–Shultz [AS03], Theorem 1.23): for product states $\omega_B \otimes \omega_M$, the evaluation $\omega_B(a \cdot c)\, \omega_M(b \cdot d) = 0$ for all ω_B, ω_M forces either $a \cdot c = 0$ or $b \cdot d = 0$. By the factor-level S4, the corresponding reverse products vanish, giving $(c \otimes d) \cdot (a \otimes b) = 0$. □

Remark 5.4 (Bootstrapping order)*.* The verification of S1–S7 on the composite avoids a potential circularity. The product-form sequential product (15) is first defined on the algebraic tensor product $V_B \otimes V_M$ (not on all of V_{BM}) by bilinear extension. Axioms S1–S7 are verified on this subspace using only the factor-level axioms and bilinearity—no reference to the full composite V_{BM} is needed. By Theorem 4.5, the algebraic tensor product equipped with this product carries EJA structure. State separation (Theorem 5.8) then shows that the product effects $\{a_i \otimes b_j\}$ are linearly independent, giving $\dim(V_{BM}) \geq \dim(V_B) \cdot \dim(V_M)$. Minimality (Definition 2.6(iii)) supplies the upper bound, yielding $V_{BM} = V_B \otimes V_M$.

5.3 Local Tomography

The key step is showing that the composite is locally tomographic: product measurements suffice to determine all joint states. Conceptually, faithful

self-modeling ensures that body–model correlations carry full information (Proposition 5.6 below); the formal proof then uses state separation and minimality.
By Definition 2.6(iv), V has no nontrivial direct-sum decomposition into order-unit subspaces. Equivalently, once the EJA structure is established, V is simple. This clause is needed for the non-degeneracy of the trace form below.

Definition 5.5 (Correlation bilinear form)**.** Let τ denote the normalized trace functional on the EJA V (Faraut–Korányi [FK94], Ch. III) and let $\circ$ denote the Jordan product. Define the *correlation bilinear form*

$$B(a,b) \;=\; \tau\big(a \circ \varphi^{-1}(b)\big)\,, \tag{16}$$

where $\varphi^{-1}\colon V_M \to V_B$ is the inverse of the tracking isomorphism.

Proposition 5.6 (Non-degeneracy)**.** *On a simple EJA, the bilinear form $B(a,b)$ is non-degenerate.*

Proof. On a simple EJA, the trace form $(a,c) \mapsto \tau(a \circ c)$ is non-degenerate (Faraut–Korányi [FK94], Proposition III.4.2). Since $\varphi\colon V_B \to V_M$ is an isomorphism (faithful self-modeling), φ^{-1} is a bijection, so $B(a,b) = \tau(a \circ \varphi^{-1}(b))$ is non-degenerate: if $B(a,b) = 0$ for all a, then $\varphi^{-1}(b) = 0$, hence $b = 0$. □

Remark 5.7 (Role of the correlation form)*.* Non-degeneracy of B provides intuition: every model effect b is correlated with some body effect a through the self-modeling isomorphism φ, so no body–model correlation is invisible to product-form observations. The formal proof of local tomography below does not use B directly; it proceeds via state separation (lower bound) and minimality (upper bound). The correlation form serves as a consistency check that the self-modeling isomorphism supplies enough correlations for product measurements to be tomographically complete, but it is not a step in the proof.

Theorem 5.8 (Local tomography)**.** *Under conditions (iii) and (iv) of Definition 2.6,*

$$\dim(V_{BM}) \;=\; \dim(V_B)\cdot\dim(V_M).$$

Proof. Lower bound. Let $\{a_1,\dots,a_d\}$ and $\{b_1,\dots,b_d\}$ be bases for V_B and V_M respectively ($d = \dim V_B = \dim V_M$). We show that the d^2 product effects $\{a_i \otimes b_j\}$ are linearly independent in V_{BM}.

Suppose $\sum_{i,j} \alpha_{ij}\, a_i \otimes b_j = 0$ in V_{BM}. By axiom (C3), every product state $\omega_B \otimes \omega_M$ belongs to the state space of V_{BM}. Evaluating:

$$0 = (\omega_B \otimes \omega_M)\Big(\sum_{i,j} \alpha_{ij}\, a_i \otimes b_j\Big) = \sum_{i,j} \alpha_{ij}\, \omega_B(a_i)\, \omega_M(b_j)$$

for all states ω_B on V_B and ω_M on V_M. Fix ω_B. Then for all ω_M:

$$\omega_M\Big(\sum_j c_j\, b_j\Big) = 0, \qquad c_j \coloneqq \sum_i \alpha_{ij}\, \omega_B(a_i).$$

States separate points in finite-dimensional order unit spaces [AS03], so $\sum_j c_j\, b_j = 0$. Since $\{b_j\}$ is a basis, $c_j = 0$ for every j, i.e.,

$$\omega_B\Big(\sum_i \alpha_{ij}\, a_i\Big) = 0 \qquad \text{for all } \omega_B \text{ and all } j.$$

State separation again gives $\sum_i \alpha_{ij}\, a_i = 0$ for every j, and since $\{a_i\}$ is a basis, $\alpha_{ij} = 0$ for all i, j. Hence the product effects are linearly independent and $\dim(V_{BM}) \geq d^2$.

Upper bound. Let $W \subseteq V_{BM}$ be the d^2-dimensional span of the product effects $\{a_i \otimes b_j\}$. The subspace W inherits OUS structure from V_{BM}: it contains the order unit $\mathbb{1}_B \otimes \mathbb{1}_M$ (since $\mathbb{1}_B \in \operatorname{span}\{a_i\}$ and $\mathbb{1}_M \in \operatorname{span}\{b_j\}$), and the induced cone $W^+ = W \cap V_{BM}^+$ is proper (the product cone is contained in W^+ and generates W). We check that W, equipped with this induced order and the product-form sequential product, satisfies (C1)–(C4): product effects lie in W by construction (C2); product states are positive functionals on W (C3, by restriction); non-signaling holds because marginals factor through V_B and V_M (C4); and the product-form SP maps W into W since $(a \otimes b) \cdot (c \otimes d) = (a \cdot c) \otimes (b \cdot d)$ is again a product effect (C1). By minimality (Definition 2.6(iii)), V_{BM} is the smallest OUS satisfying these axioms. Since W satisfies them and has dimension d^2, we have $\dim(V_{BM}) \leq d^2$. □

5.4 The Entangled Sector

The *entangled sector* consists of states in V_{BM} not reachable as convex combinations of product states. Minimality (Definition 2.6(iii)) eliminates this sector by construction: V_{BM} is defined as the smallest OUS satisfying the composite axioms, so it contains no "hidden" entangled structure beyond what the sequential product and non-signaling constraints require.

For complex quantum mechanics ($V = M_n(\mathbb{C})^{\mathrm{sa}}$), the minimal and maximal composites coincide [BUvdW23]: the standard Hilbert space tensor

product $M_{nm}(\mathbb{C})^{\mathrm{sa}}$ is the unique composite satisfying (C1)–(C4). This is why minimality is not a restrictive assumption for the complex case.

For real or quaternionic quantum mechanics, the minimal and maximal composites differ. This discrepancy manifests as a failure of local tomography and is the mechanism by which these types are excluded:

Remark 5.9 (Negative checks). Dimension counting reveals that local tomography fails for all non-complex types at $n = 2$:

Type	$d = \dim(V)$	d^2	dim(composite)	Local tomography?
$M_2(\mathbb{R})^{\mathrm{sa}}$	3	9	10	No $(9 < 10)$
$M_2(\mathbb{C})^{\mathrm{sa}}$	4	16	16	Yes $(16 = 16)$
$M_2(\mathbb{H})^{\mathrm{sa}}$	6	36	28	No $(36 > 28)$

The real composite is "too large" (product measurements cannot distinguish all joint states), while the quaternionic composite is "too small" (product measurements are linearly dependent on the joint system). Only the complex type achieves the exact match $d^2 = \dim(\text{composite})$.

6 Type Exclusion and C^*-Algebra Promotion

6.1 The Jordan–von Neumann–Wigner Classification

By Theorem 4.5, the state space V is a Euclidean Jordan algebra (EJA). The Jordan–von Neumann–Wigner classification [JvNW34] asserts that every finite-dimensional simple EJA belongs to one of five families:

Type	Notation	dim	LT?	Exclusion mechanism
Real	$M_n(\mathbb{R})^{\mathrm{sa}}$	$n(n{+}1)/2$	Only $n{=}1$	Dimension mismatch
Complex	$M_n(\mathbb{C})^{\mathrm{sa}}$	n^2	All n	—
Quaternionic	$M_n(\mathbb{H})^{\mathrm{sa}}$	$n(2n{-}1)$	Only $n{=}1$	Dimension mismatch
Spin factor	V_n $(n \geq 3)$	$n{+}1$	Only V_3	Barnum–Wilce
Albert	$M_3(\mathbb{O})^{\mathrm{sa}}$	27	Never	No composite exists

The spin factors overlap with the matrix types at low dimensions: $V_2 \cong M_2(\mathbb{R})^{\mathrm{sa}}$, $V_3 \cong M_2(\mathbb{C})^{\mathrm{sa}}$, and $V_5 \cong M_2(\mathbb{H})^{\mathrm{sa}}$.

6.2 Excluding Non-Complex Types

We now show that local tomography (Theorem 5.8) excludes every non-complex type.

Real matrices, $M_n(\mathbb{R})^{\mathrm{sa}}$ ($n \geq 2$). The local tomography condition $[\dim(V)]^2 = \dim(\text{composite})$ becomes

$$\left[\tfrac{n(n+1)}{2}\right]^2 = \tfrac{n^2(n^2+1)}{2}\,.$$

Simplifying: $(n+1)^2 = 2(n^2+1)$, which reduces to $(n-1)^2 = 0$, forcing $n = 1$ (trivial, one-dimensional). For $n \geq 2$, the real composite is strictly larger than the product-effect span: $d^2 < \dim(\text{composite})$, and product measurements leave an entangled sector unresolved [BW14].

Quaternionic matrices, $M_n(\mathbb{H})^{\mathrm{sa}}$ ($n \geq 2$). The analogous condition yields $[n(2n-1)]^2 = n^2(2n^2-1)$, which again reduces to $(n-1)^2 = 0$, forcing $n = 1$ (trivial). For $n \geq 2$, the quaternionic composite has *fewer* dimensions than d^2: product effects are linearly dependent on the joint system [BW14].

Spin factors, V_n ($n \geq 4$). The cross-identifications $V_2 \cong M_2(\mathbb{R})^{\mathrm{sa}}$ and $V_5 \cong M_2(\mathbb{H})^{\mathrm{sa}}$ are already excluded above. For V_n with $n \geq 4$ and $n \neq 5$, these are "pure" spin factors not isomorphic to any $M_k(\mathbb{K})^{\mathrm{sa}}$. Barnum and Wilce [BW14] prove that no locally tomographic composite exists for these types (Theorem 4.1 of [BW14]).

Albert algebra, $M_3(\mathbb{O})^{\mathrm{sa}}$. The exclusion here is the strongest: Barnum, Graydon, and Wilce [BGW20] prove that no non-signaling composite with product states exists for any EJA containing an Albert algebra summand (Theorem 1 of [BGW20]). The 27-dimensional exceptional Jordan algebra cannot form *any* consistent bipartite system, even a non-locally-tomographic one.

Complex matrices, $M_n(\mathbb{C})^{\mathrm{sa}}$ ($n \geq 1$). $\dim(M_n(\mathbb{C})^{\mathrm{sa}}) = n^2$, and the composite $M_{n^2}(\mathbb{C})^{\mathrm{sa}}$ has dimension $(n^2)^2 = n^4 = (n^2)^2$. Local tomography holds for all n: the dimension formula $d = n^2$ is *multiplicatively stable*. The complex type is the unique survivor.

6.3 C*-Algebra Promotion

The surviving EJA is promoted to a C*-algebra by a single published theorem.

Theorem 6.1 (C*-algebra structure; [vdW19b, Theorem 3])**.** *Let V be a sequential product space satisfying S1–S7. If the composite $V \otimes V$, equipped with the product-form sequential product, is also a sequential product space and the composite is locally tomographic, then V is the self-adjoint part of a C*-algebra.*

Hypothesis	**Status**	**Established in**
V satisfies S1–S7	Verified	Section 4
$V \otimes V$ with product-form SP satisfies S1–S7	Verified	Proposition 5.3
Composite is locally tomographic	Verified	Theorem 5.8

Theorem 6.1 combined with the type exclusion of Section 6.2 establishes that $V \cong M_n(\mathbb{C})^{\mathrm{sa}}$ for some $n \geq 2$. The case $n = 1$ ($V \cong \mathbb{R}$) is excluded by the nontriviality condition in Definition 2.6(i): a one-dimensional OUS has no nontrivial projective units, so the Peirce structure is vacuous and the self-modeling cycle produces no nontrivial sequential product. This is the self-adjoint part of the C*-algebra $M_n(\mathbb{C})$, and the C*-promotion is complete.

Remark 6.2 (Consistency with independent results)*.* Two further published theorems provide independent consistency checks on the conclusion $V \cong M_n(\mathbb{C})^{\mathrm{sa}}$.

Barnum–Wilce [BW14]. Barnum and Wilce prove that in a locally tomographic dagger-compact probabilistic theory containing a qubit, every EJA system type is $M_n(\mathbb{C})^{\mathrm{sa}}$. Their hypotheses are categorical (they require a full monoidal theory with dagger structure on every object, not merely one composite), and we do not construct this categorical apparatus here. However, our conclusion is consistent with their result: after establishing $V = M_n(\mathbb{C})^{\mathrm{sa}}$ via Theorem 6.1, the category of complex matrix algebras with Hilbert space tensor products does form a locally tomographic dagger-compact theory.

Hanche-Olsen [HO85]. Hanche-Olsen proves that a JB-algebra E whose tensor product $E \otimes E_2$ with the qubit algebra $E_2 = M_2(\mathbb{C})^{\mathrm{sa}}$ is again a JB-algebra (with factorization identities for the Jordan product) must be the self-adjoint part of a C*-algebra. The qubit subsystem E_2 is available only *after* type exclusion has already established the complex type, so this theorem does not provide an independent route to the conclusion. It does, however, confirm the C*-algebra structure through entirely different mathematical machinery (JB-algebra tensor products rather than sequential product classification).

Neither of these two broader theorems is used in the derivation (unlike the specific dimension-counting and no-composite results of [BW14] and [BGW20] invoked in Section 6.2 above); both serve as independent cross-checks that the conclusion is robust.

6.4 The Involution

Theorem 6.1 and the type exclusion of Section 6.2 establish that $V = M_n(\mathbb{C})^{\mathrm{sa}}$ is the self-adjoint part of the C*-algebra $A = M_n(\mathbb{C})$. The C*-algebra involution is the conjugate transpose:

$$X^* \;=\; X^\dagger \;=\; \overline{X}^T.$$

This involution satisfies four defining properties:

(P1) **Involutive:** $(X^*)^* = X$.

(P2) **Anti-multiplicative:** $(XY)^* = Y^*X^*$.

(P3) **Fixed-point set:** $\{X \in M_n(\mathbb{C}) : X^* = X\} = M_n(\mathbb{C})^{\mathrm{sa}}$.

(P4) **C*-identity:** $\|X^*X\| = \|X\|^2$ (operator norm).

Property (P3) identifies the self-adjoint part recovered from the self-modeling construction with the standard notion of Hermitian matrices. Property (P4) equips $M_n(\mathbb{C})$ with its C*-norm, completing the C*-algebra structure.

6.5 Main Theorem

We can now state the central result.

Theorem 6.3 (Self-modeling implies complex quantum mechanics)**.** *Let (V, φ, V_{BM}) be a self-modeling system (Definition 2.6). Then:*

1. *$V \cong M_n(\mathbb{C})^{\mathrm{sa}}$ for some $n \geq 2$;*
2. *the sequential product is the Lüders product $a \cdot b = \sqrt{a}\, b \sqrt{a}$;*
3. *$M_n(\mathbb{C})$ carries the conjugate-transpose involution $X^* = X^\dagger$.*

In particular, the complex field, the Jordan product, and the C*-involution are all consequences of the self-modeling definition (Definition 2.6). Local tomography follows from the internal-accessibility condition (iii); the complex field and C*-structure then follow from the classification machinery.

Proof outline.

1. The self-modeling constraint yields a sequential product on V via the "test–update–test" operational semantics (Section 3).
2. The corrected product formula (12), incorporating the Peirce 1-space feedback selected by faithful tracking, satisfies S1–S7 (Section 4).
3. By Theorem 4.5 (van de Wetering), V is an EJA.
4. The composite V_{BM} with the product-form SP inherits S1–S7 (Proposition 5.3), and state separation combined with minimality yields local tomography (Theorem 5.8).
5. Dimension counting excludes real and quaternionic types; published no-composite results exclude spin factors ([BW14], Thm. 4.1) and the Albert algebra ([BGW20], Thm. 1) (Section 6.2).
6. Theorem 6.1 (C*-promotion via van de Wetering) yields $V = M_n(\mathbb{C})^{\mathrm{sa}}$ with the conjugate-transpose involution. □

□

7 Discussion

7.1 Summary of Results

Self-modeling systems, as formalized in Definition 2.6, are exactly the complex matrix algebras $M_n(\mathbb{C})^{\mathrm{sa}}$ ($n \geq 2$) equipped with the Lüders sequential product $a \cdot b = \sqrt{a}\, b \sqrt{a}$ and the conjugate-transpose involution $X^* = X^\dagger$. The identification proceeds through a chain of eight steps (see Figure 1): the self-modeling definition entails a sequential product on the effect space; the product satisfies the seven axioms of van de Wetering, establishing Euclidean Jordan algebra structure; only complex systems admit the minimal composite the definition requires; and the surviving algebra is a C^*-algebra via published theorems of van de Wetering [vdW19b], Barnum–Wilce [BW14], and Hanche-Olsen [HO85].

The complex field, the Jordan product, and the C^*-involution are consequences of Definition 2.6. Local tomography follows from Condition (iii) (minimal composite) via state separation (lower bound) and minimality (upper bound); the complex field and C^*-structure then follow from the classification machinery. None are independent postulates. The four conditions of the definition each unpack one word of "self-modeling system"; we

now examine their individual roles, restrictiveness, and the consequences of weakening them.

Dependency audit. We itemize explicitly what each ingredient contributes to the conclusion, to prevent blurring of the derivation structure.

- **Ambient framework** (spectral OUS with Alfsen–Shultz compressions and Peirce decomposition): provides the setting in which both classical and quantum theories live; supplies spectral decomposition, faces, compressions, and the Peirce direct sum. This is rich structure but is *not biased toward quantum mechanics* (Remark 7.8).
- **Definition 2.6(i)–(ii)** (spectral OUS + faithful tracking): provides the operational product $p \cdot q = C_p(q)$ on sharp effects and the Peirce 1-space ansatz (7); constrains the mixing function to $f = \sqrt{\lambda_i \lambda_j}$ via S5 (Proposition 3.3). This is the novel construction of the paper.
- **Published theorem** (van de Wetering [vdW19b], Theorem 1): converts S1–S7 into EJA structure.
- **Definition 2.6(iii)–(iv)** (minimal composite + simplicity): yields local tomography (Theorem 5.8), which excludes all non-complex simple EJA types.
- **Published theorems** (van de Wetering [vdW19b] Theorem 3, Barnum–Wilce [BW14] Theorem 4.1, BGW [BGW20] Theorem 1): promote the surviving EJA to C^*-algebra structure and exclude spin factors and the Albert algebra.

The novel mathematics of the paper is concentrated in the second item: the self-modeling sequential product construction and the proof that S5 forces scalar Peirce 1-space action (Proposition 3.3). Everything upstream is framework; everything downstream is published classification machinery.

7.2 The Four Conditions

We analyze each condition's role, restrictiveness, and the consequences of weakening it.

Condition (i): Finite-dimensional spectral OUS. We argue in Section 7.3 that this follows from finite capacity of the embedding region. It is the most restrictive assumption in the formal sense. The spectral decomposition $a = \sum_i \lambda_i p_i$, which underlies the entire product construction (7), requires a well-behaved spectral theory. In finite dimensions, this is guaranteed by the Alfsen–Shultz theory [AS03]. In infinite dimensions, spectral theory introduces domain issues: the spectral measure replaces the fi-

nite sum, unbounded effects arise, and the Peirce decomposition becomes a conditional-expectation-valued integral. We expect the core argument to extend to type I factors (direct integrals of matrix algebras), where the spectral decomposition retains its essential features. Extension to type II or III von Neumann algebras would require substantially new techniques, particularly for the Peirce feedback construction and the facial orthogonality argument for (S4).

Condition (ii): Faithful self-model. Faithfulness means $\varphi : V_B \to V_M$ is an order isomorphism—the model is a perfect internal copy. As discussed in Remark 3.5, the mixing function $f = \sqrt{\lambda_i \lambda_j}$ is determined by internal consistency ((S5)), not directly by φ. Heuristically, $f = 0$ corresponds to a trivial (non-faithful) model and $f = \sqrt{\lambda_i \lambda_j}$ to a faithful one, but we do not prove a quantitative theorem linking the fidelity of φ to the value of f. Whether approximate self-models (where φ is injective but not surjective) yield "partially coherent" products with $|f| < \sqrt{\lambda_i \lambda_j}$ is a conjectural extension that we leave to future work.

Condition (iii): Minimal composite. We argue in Section 7.3 that minimality is the natural composite for a maximally efficient self-modeler. The body–model composite V_{BM} is defined as the minimal order unit space carrying product states, product effects, non-signaling constraints, and a product-form sequential product. For complex quantum mechanics, the minimal and maximal composites coincide: the Hilbert space tensor product $M_{nm}(\mathbb{C})^{\mathrm{sa}}$ is the unique composite satisfying (C1)–(C4) [BUvdW23]. This is a feature of the complex case, not a coincidence: it means Condition (iii) is not restrictive for the theory we derive. For a priori exploration, one could ask whether maximal composites yield different conclusions. For the real and quaternionic cases, minimal $\neq$ maximal: the difference manifests as a failure of local tomography (the real composite has "hidden entangled states" that product measurements cannot resolve). For the non-complex types, the discrepancy between minimal and maximal composites is exactly the opacity that Section 7.3 rules out on self-modeling grounds.

Condition (iv): Simplicity. We argue in Section 7.3 that simplicity corresponds to "whole models whole"—the self-modeler models all of itself, not independent parts. The trace form non-degeneracy argument in Proposition 5.6 requires the EJA to be simple—having no nontrivial Jordan ideal. An EJA with a direct sum decomposition $V = V_1 \oplus V_2$ splits into superse-

lection sectors invisible to each other; a system in one sector cannot model another, so the self-model does not cover the whole. However, each simple summand satisfies our theorem independently: if V_1 and V_2 each admit faithful self-models, they are each $M_{n_i}(\mathbb{C})^{\mathrm{sa}}$ by the main theorem applied to each summand. The direct sum then describes a system with superselection sectors, each governed by complex quantum mechanics. Thus Condition (iv) restricts to irreducible systems, which is the generic case.

7.3 The Definition Revisited

Definition 2.6 formalizes self-modeling as four conditions. Section 2.4 argued that each unpacks one word of "self-modeling system." We now examine why each clause is forced—and why removing any one would break the concept.

Finite capacity. A self-modeler that contains an isomorphic copy of itself within itself must have finite state-space dimension. An infinite-dimensional system would require an infinite-dimensional internal model, but a subsystem embedded in a finite spatial region has finite information capacity [Bek81]. The spectral structure (distinguishable states, resolutions of unity) is not an abstract regularity condition: it is the minimal apparatus for *probing*—a system that tests its own state requires states to distinguish. In this sense, Condition (i) is not a restriction on the formalism but a structural consequence of what "can probe itself in a finite region" means operationally.

Faithfulness. This is the operational premise itself. No further argument is needed.

Internality. Condition (iii) (minimal composite) says the body–model composite carries no structure beyond what product measurements can resolve. The argument is not efficiency but *accessibility*.

A self-modeler's only interface with its own body–model composite is through product measurements: test an effect on the body, test an effect on the model, combine the outcomes. If the composite carried states that product measurements could not distinguish—an entangled sector invisible to product observations—those states would be aspects of the system's own composite that it cannot probe. A system with such opaque internal structure is not modeling *itself*; it is modeling a shadow of itself, blind to correlations it cannot access.

In the non-complex EJA types, this opacity is exactly what occurs. For real quantum mechanics ($n \geq 2$), the maximal composite has dim $> d^2$: there are body–model states that product measurements cannot resolve, giving the self-model blind spots [BW14]. For quaternionic systems, dim $< d^2$: product effects are linearly dependent, making the self-model's observations redundant. In both cases, the self-modeler fails to faithfully track its own composite state. Only in the complex case do the minimal and maximal composites coincide [BUvdW23], meaning product measurements see *everything*—the composite has no hidden structure, and the self-model is genuinely faithful at the composite level.

Minimality is therefore not an independent design choice imposed from outside. It is forced by the conjunction of faithfulness ((ii)) and internality: if the only available measurements are product measurements, and the model must faithfully capture all accessible structure, then the composite can contain nothing that product measurements cannot resolve.

Irreducibility. Condition (iv) (simple EJA) says the algebra has no nontrivial ideal—it cannot be decomposed into independent blocks. A self-modeler that decomposes into independent subsystems is not modeling *all* of itself: each block models only itself, not the whole. Simplicity is the algebraic expression of "whole models whole." As noted in Section 7.2, each simple summand independently satisfies the main theorem, so the assumption restricts to irreducible systems rather than excluding composite ones.

The mathematical theorem requires the four conditions as stated. The conceptual argument is that they are not independent postulates but the structural shape that self-modeling takes in a finite-capacity system—the minimum a self-modeler could be. The following characterization makes the relationship precise.

Corollary 7.1 (Characterization)**.** *Among simple finite-dimensional EJAs, a type admits a faithful self-model whose minimal composite is locally tomographic if and only if it is $M_n(\mathbb{C})^{\mathrm{sa}}$ for some $n \geq 2$.*

Proof. The forward direction is the main theorem. For the converse, take $V_M = V_B = M_n(\mathbb{C})^{\mathrm{sa}}$ with $\varphi = \mathrm{id}$. The minimal composite of $M_n(\mathbb{C})^{\mathrm{sa}}$ with itself is $M_{n^2}(\mathbb{C})^{\mathrm{sa}}$, which satisfies dim $= n^4 = (n^2)^2 = [\dim(V_B)]^2$. Local tomography holds, and the product-form sequential product inherits S1–S7 from the factors. □

Remark 7.2 (Why the other types fail)*.* The converse fails for every non-complex simple EJA type. For real matrices ($n \geq 2$), the minimal composite

has dim $> [\dim(V)]^2$: product measurements leave an entangled sector unresolved, so the self-model has *blind spots* about body–model correlations. For quaternionic matrices ($n \geq 2$), dim $< [\dim(V)]^2$: product effects are linearly dependent, so the self-model's observations are *redundant*. For spin factors and the Albert algebra, no locally tomographic composite exists at all. Complex quantum mechanics is the unique theory in which a finite-dimensional system can have complete self-knowledge through product measurements on the body–model composite.

Corollary 7.1 says that the conditions of our theorem are not merely sufficient for complex quantum mechanics—they *characterize* it. There is no simple finite-dimensional EJA satisfying our conditions that is not complex QM, and no complex QM system that fails to satisfy them. The "axiomatic corridor" identified by the theorem is, in this precise sense, the corridor of self-modeling itself.

7.4 Comparison with Other Programs

Table 1 provides a summary comparison. We now discuss several programs in more depth.

Masanes–Müller (2011–2013). The Masanes–Müller program [MM11, MMAPG13] achieves the closest rival in parsimony, using three postulates: the existence of an information unit, continuous reversibility, and tomographic locality. The key difference is that their tomographic locality postulate assumes local tomography (effectively selecting the complex field as input), whereas we *derive* local tomography from faithful tracking via the EJA trace form non-degeneracy argument (Theorem 5.8). Their continuous reversibility axiom—that every reversible transformation can be continuously connected to the identity—has no analogue in our framework; instead, the reversibility structure emerges from the C^*-algebra promotion. On the other hand, their framework handles infinite dimensions more naturally than ours, and their postulates have a cleaner operational interpretation in terms of information processing.

Chiribella–D'Ariano–Perinotti (2011). The CDP program [CDP11] uses six axioms, with Purification as the central workhorse: every mixed state of a system arises as the marginal of a pure state of a larger system. Purification is a powerful constraint that immediately selects complex quantum mechanics from the landscape of GPTs. We derive the C^*-algebraic

structure without assuming purification; purification then follows as a *theorem* of $M_n(\mathbb{C})^{\mathrm{sa}}$, not a postulate. However, the CDP framework provides a complete categorical description of quantum theory (including channels, instruments, and higher-order maps) that goes well beyond our algebraic result.

Van de Wetering's effectus theory (2019). Our work relies heavily on van de Wetering's theorems [vdW19b], but the relationship to his effectus-theoretic reconstruction [vdW19a] is complementary rather than subsumptive. The effectus approach uses approximately five axioms on the categorical structure of a theory (effect algebra homomorphisms, images, compressions, quotients, sharp effects). Our approach is more operational: we replace categorical axioms with the self-modeling definition and derive the sequential product structure that feeds into van de Wetering's classification theorems. The two programs share a common downstream engine (Theorems 1 and 3 of [vdW19b]) but differ in what they take as input.

Alfsen–Shultz geometry. The Alfsen–Shultz program [AS03] characterizes C^*-algebras via the geometry of state spaces, using the concept of an *orientation* on the state space. Their characterization is geometrically elegant: a JB-algebra is the self-adjoint part of a C^*-algebra if and only if its state space admits a consistent orientation (a choice of "which way around" each face). In our framework, the orientation is not an input but emerges from the self-modeling construction: the tracking map φ provides a canonical correspondence between faces of V_B and faces of V_M that implicitly defines an orientation. Making this connection precise is a direction for future investigation.

7.5 Future Directions

Infinite-dimensional extension. The most important open problem is extending the result to infinite-dimensional systems. The key obstacles are:

(i) the spectral decomposition becomes a spectral measure, and the Peirce feedback formula (7) becomes an operator-valued integral;

(ii) the facial orthogonality argument for (S4) relies on Alfsen–Shultz propositions that may require modification in infinite dimensions;

(iii) the minimal composite construction must be replaced by an appropriate operator-algebraic tensor product.

We conjecture that the result extends to type I factors, and that the self-modeling condition for type III algebras is vacuous (no faithful finite-dimensional self-model exists).

Approximate self-models and decoherence. If the tracking map φ is only approximately faithful (e.g., $\|\varphi(a) - a\| \leq \varepsilon\|a\|$ for small ε), the mixing function f would not saturate the positivity bound. The resulting "partially coherent" sequential product interpolates between classical ($f = 0$, full decoherence) and quantum ($f = \sqrt{\lambda_i \lambda_j}$, full coherence). This suggests a connection between self-modeling fidelity and decoherence: a system that imperfectly models itself would exhibit intrinsic decoherence proportional to the modeling error. Formalizing this connection could provide an operational account of the quantum-to-classical transition.

Categorical formulation. The self-modeling construction could be formulated in the language of effectus theory or process theories [SSC21], where the tracking map φ becomes a morphism in a suitable category and the product construction becomes a functor. This would clarify the relationship to the categorical reconstruction programs and might reveal additional structure (e.g., whether self-modeling systems form a subcategory of effectus algebras with special properties).

Connection to stochastic–quantum bijection. Barandes [Bar25] has established a bijection between stochastic processes and quantum dynamics, showing that unitary quantum evolution can be recast as a stochastic process with indivisibility constraints. If a faithful self-model generates a stochastic process (via the sequential test–update–test cycle), the Barandes bijection would map it to a quantum channel. Investigating whether the self-modeling premise can be rephrased in stochastic terms—and whether the Barandes bijection provides an alternative derivation route—is an intriguing direction.

Non-Markovian self-models. Our construction assumes a memoryless update cycle: testing p triggers a single compression $C_{\varphi(p)}$, with no dependence on prior measurements. A non-Markovian self-model, where the update depends on a history of prior tests, would introduce memory effects and might connect to quantum memory channels or process matrices.

7.6 Anticipated Objections

We address several objections that a careful reader might raise.

Remark 7.3 ("Isn't $\sqrt{\cdot}$ importing Hilbert space structure?"). No. The square root in the mixing function operates on *real eigenvalues* $\lambda_i, \lambda_j \in [0,1]$: it is the elementary real-valued function $\sqrt{x} : [0,1] \to [0,1]$, the unique non-negative real number whose square is x. No operator square root, no spectral theorem for self-adjoint operators on a Hilbert space, no functional calculus of C^*-algebras, and no trace operation is invoked. The distinction is between a scalar function on the reals (which we use) and an operator function on a C^*-algebra (which we do not). See Section 3.5 for a complete inventory of the mathematical objects used in the construction.

Remark 7.4 ("Isn't minimality just local tomography in disguise?"). The objection is that Condition (iii) (minimal composite) does the real work of excluding non-complex types, and that calling it "self-modeling" is relabeling. We take this objection seriously and answer it on three levels.

Conceptual. The argument of Section 7.3 (Internality) is that a self-modeler whose only interface with its own composite is product measurements *cannot* have a composite carrying states those measurements fail to resolve. Minimality follows from faithfulness ((ii)) plus internality: the composite must contain nothing the self-modeler cannot probe. This is a constraint *about self-modeling*, not a free-standing information-processing postulate.

Mathematical. For the theory we derive (complex quantum mechanics), the minimal and maximal composites coincide—this is a theorem, not an assumption [BUvdW23]. So Condition (iii) is *automatically satisfied* by every complex system: it imposes no restriction beyond what complex QM already has. It restricts only the non-complex types, where the discrepancy between minimal and maximal composites manifests as exactly the blind spots or redundancies that Section 7.3 rules out on self-modeling grounds.

Structural. Local tomography is a property of a composite ($\dim V_{BM} = (\dim V)^2$). Minimality is a construction principle for what the composite *is*. The theorem that the minimal composite of a simple EJA with itself is locally tomographic if and only if the EJA is complex is a *consequence* of the definition, not a restatement. The derivation—state separation for the lower bound, explicit construction of W for the upper bound—is given in full in Theorem 5.8.

Remark 7.5 ("Isn't φ mathematically inert?"). The compression product $p \cdot q = C_p(q)$ for sharp effects exists in any spectral OUS with compressions, independent of φ. However, compressions alone give only a partial product (defined on sharp effects); the extension to general effects requires the Peirce 1-space restoration (7), whose operational justification is the self-modeling

feedback loop (the model updates, then the body is re-tested). Without the self-modeling cycle, one has compressions but no reason to assemble them into a sequential product on all effects. The tracking map φ provides the product; the axioms pin its parameters. See Remark 3.5 for the division of labor.

Remark 7.6 ("What about infinite dimensions?"). Our result is restricted to finite-dimensional systems (Section 2.1). This is stated explicitly in the abstract, in the main theorem (Theorem 6.3), and in the discussion above. The finite-dimensional restriction is genuine, not cosmetic: the spectral decomposition, the Peirce feedback construction, and the facial orthogonality argument all rely on finite-dimensional spectral theory. We view the infinite-dimensional extension as the most important open problem; see Section 7.5.

Remark 7.7 ("How does this compare to effectus theory?"). Van de Wetering's effectus-theoretic reconstruction [vdW19a] uses categorical axioms on effect algebras to arrive at the same classification theorems we invoke. The relationship is complementary: we replace categorical axioms with the self-modeling definition (Definition 2.6) and derive the sequential product structure that effectus theory axiomatizes directly. Our approach trades categorical generality for operational specificity. Both programs ultimately rely on the same downstream theorems (Theorems 1 and 3 of [vdW19b]), so they are complementary rather than competing.

Remark 7.8 ("Isn't the spectral OUS framework already 80% quantum?"). A spectral order unit space with compressions, Peirce decomposition, and resolution of unity is rich structure—but it is not biased toward quantum mechanics. Classical probability theory lives comfortably within this framework as the special case where all Peirce 1-spaces are trivial ($V_1(p_i, p_j) = \{0\}$ for all projective units) and the sequential product reduces to pointwise multiplication on a simplex. The content of our derivation is that self-modeling *requires* non-trivial Peirce 1-space content: the off-diagonal structure that distinguishes quantum from classical. Without self-modeling, one could have a perfectly valid spectral OUS that is entirely classical. The Alfsen–Shultz framework is the minimal setting in which both classical and quantum theories can be formulated on equal footing; self-modeling *is* the quantum case.

Remark 7.9 ("Is faithful self-modeling physically motivated?"). Self-modeling is motivated by the observation that quantum systems are routinely used to model other quantum systems (quantum simulation, quantum error correction, quantum tomography). The faithfulness condition—that the model captures all operationally accessible structure—is the requirement that the simulation is *exact*, not approximate. In quantum computation, a universal

quantum computer can faithfully simulate any quantum system of equal or smaller dimension; our premise asks only that the system can simulate *itself*. Our result is a mathematical theorem about order unit spaces, independent of interpretive commitments about consciousness or cognition.

7.7 Conclusion

Faithful self-modeling is complex quantum mechanics. The four conditions of Definition 2.6—spectral state space, faithful tracking, minimal composite, and irreducibility—characterize exactly $M_n(\mathbb{C})^{\mathrm{sa}}$. Starting from these conditions, we constructed a sequential product on the effect algebra. This product satisfies van de Wetering's seven axioms, establishing Euclidean Jordan algebra structure. The self-modeling definition implies local tomography for the body–model composite, which excludes all non-complex EJA types and, via the theorems of van de Wetering, Barnum–Wilce, and Hanche-Olsen, promotes the algebra to $M_n(\mathbb{C})^{\mathrm{sa}}$ with the conjugate-transpose involution.

The complex field, the Jordan product, the C^*-involution, and local tomography are not independent postulates of quantum theory. They are consequences of the self-modeling definition (Definition 2.6). The four conditions of the definition—finite dimensionality, faithfulness, minimal composite, and simplicity—unpack four words of "self-modeling system" and constitute what self-modeling *is* in the finite-dimensional OUS setting.

Machine verification. The derivation chain is formalized in Lean 4 with Mathlib (v4.28.0). The formalization comprises 10 files (∼2,400 lines) and produces zero `sorry` declarations. Four axioms are used: (1) the self-modeling bridge, which encapsulates the sequential product construction of Sections 3–4; (2) the spectral Jordan identity (van de Wetering 2019, §4); (3) local tomography (Definition 2.6(iii)—the minimality-to-dimension argument of Theorem 5.8); and (4) the C^*-promotion theorem (van de Wetering 2019, Theorem 3). Axioms (2)–(4) are published theorems; axiom (1) bridges the paper's novel construction to the formalized framework. Two concrete models—diagonal 2×2 matrices and the spin factor—satisfy all S1–S7 from scratch, verifying that the axiom system is consistent. The formalization is available at `https://github.com/ehrlich-b/research/tree/jmp-submission-2026-03-28/lean`.

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A Detailed Proofs

A.1 Proof of (S4) (Orthogonality Symmetry)

We give the full proof that the self-modeling sequential product (12) satisfies axiom (S4): if $a \cdot b = 0$ then $b \cdot a = 0$. The proof depends only on the facial geometry of spectral order unit spaces and the vanishing of the mixing function at zero eigenvalues ($f(0, x) = 0$), making it independent of the specific choice $f = \sqrt{\lambda_i \lambda_j}$.

Theorem A.1. *Let $(V, \leq, \nVdash)$ be a finite-dimensional spectral order unit space equipped with the sequential product* (12)*. If $a \cdot b = 0$ for effects $a, b \in [0, 1]_V$, then $b \cdot a = 0$.*

Proof. Write the spectral decomposition $a = \sum_{i=1}^{n} \lambda_i \, p_i$, with $\lambda_i \geq 0$, the p_i mutually orthogonal projective units, and $\sum_i p_i = \nVdash$. The condition $a \cdot b = 0$ reads

$$\sum_i \lambda_i \, C_{p_i}(b) + \sum_{i<j} f(\lambda_i, \lambda_j) \, P_{ij}(b) = 0. \tag{17}$$

By the Peirce decomposition (3), the terms $C_{p_i}(b) \in V_2(p_i)$ and $P_{ij}(b) \in V_1(p_i, p_j)$ lie in mutually orthogonal subspaces. In a direct sum, a sum vanishes if and only if each summand vanishes. Therefore (17) implies

$$\lambda_i \, C_{p_i}(b) = 0 \quad \text{for each } i, \tag{18}$$

$$f(\lambda_i, \lambda_j) \, P_{ij}(b) = 0 \quad \text{for each } i < j. \tag{19}$$

Case A: Full-rank ($\lambda_i > 0$ for all i). From (18), $C_{p_i}(b) = 0$ for all i. From (19), since $f(\lambda_i, \lambda_j) > 0$ when both arguments are positive (this holds

for any mixing function with $f > 0$ on $(0,1]^2$, including $f = \sqrt{\lambda_i \lambda_j}$), we get $P_{ij}(b) = 0$ for all $i < j$. The completeness of the Peirce decomposition (3) then gives

$$b = \sum_i C_{p_i}(b) + \sum_{i<j} P_{ij}(b) = 0.$$

Hence $b \cdot a = 0 \cdot a = 0$ (by (S1), the map $b \mapsto b \cdot a$ sends 0 to 0).

Case B: Rank-deficient ($\lambda_k = 0$ for some k). Partition the indices into $I_+ = \{i : \lambda_i > 0\}$ and $I_0 = \{i : \lambda_i = 0\}$, both nonempty. From (18): $C_{p_i}(b) = 0$ for all $i \in I_+$. From (19): for $i, j \in I_+$, $f(\lambda_i, \lambda_j) > 0$ gives $P_{ij}(b) = 0$; for $i \in I_+$, $j \in I_0$, $f(\lambda_i, 0) = 0$ gives no constraint on $P_{ij}(b)$.

The key step is the *facial absorption theorem* (Alfsen–Shultz [AS03], Proposition 7.43):

> If $C_p(b) = 0$ and $b \geq 0$, then $b \in \mathrm{face}(p^\perp)$, the face generated by $p^\perp = \mathbb{1} - p$.

Define $p_+ = \sum_{i \in I_+} p_i$ (the support projection of a). Since $C_{p_i}(b) = 0$ for each $i \in I_+$, and compressions for orthogonal projective units act independently on their respective faces, we have $C_{p_+}(b) = \sum_{i \in I_+} C_{p_i}(b) = 0$. By facial absorption, $b \in \mathrm{face}(p_+^\perp)$, where $p_+^\perp = \sum_{i \in I_0} p_i$.

This means b is supported entirely on the zero-eigenvalue block of a. For the reverse product $b \cdot a$, write $b = \sum_j \mu_j \, q_j$ (spectral decomposition of b). Since $b \in \mathrm{face}(p_+^\perp)$, each spectral projector q_j with $\mu_j > 0$ satisfies $q_j \leq p_+^\perp$ in the face lattice.

Now consider $b \cdot a$:

$$b \cdot a = \sum_j \mu_j \, C_{q_j}(a) + \sum_{j<k} f(\mu_j, \mu_k) \, Q_{jk}(a),$$

where Q_{jk} denotes the Peirce projection for the q_j-basis. For each j with $\mu_j > 0$: since $q_j \leq p_+^\perp$ and a is supported on p_+ (all nonzero eigenvalues correspond to p_+), the facial orthogonality of complementary faces gives $C_{q_j}(a) = 0$. The Peirce 1-space terms $Q_{jk}(a)$ either vanish by facial structure (when both $\mu_j, \mu_k > 0$, so both $q_j, q_k \leq p_+^\perp$, and a has no component in $V_1(q_j, q_k)$ since a is supported on the complementary face) or carry zero weight $f(\mu_j, 0) = 0$ (when $\mu_k = 0$). Hence $b \cdot a = 0$.

Case $a = 0$. Trivial: $0 \cdot b = 0$ implies nothing, and $b \cdot 0 = 0$ by (S1). □

Corollary A.2 (φ-independence of (S4))**.** *Axiom* (S4) *holds for the sequential product* (7) *with* any *mixing function f satisfying $f(0, x) = 0$ for all $x \in [0,1]$, not only for the faithful choice $f = \sqrt{\lambda_i \lambda_j}$.*

Proof. The proof of Theorem A.1 uses only: (i) the Peirce direct sum structure, (ii) the vanishing $f(\lambda, 0) = 0$, and (iii) facial absorption. The specific form of f for positive arguments appears only in Case A (where $f > 0$ on $(0, 1]^2$ suffices) and in the $(i, j \in I_+)$ part of Case B (same condition). Any mixing function with $f(0, x) = 0$ and $f > 0$ on $(0, 1]^2$ satisfies the proof's requirements. □

A.2 Proof of Local Tomography (Trace Form Non-Degeneracy)

We give the detailed proof that the non-degeneracy of the EJA trace form, combined with faithful self-modeling and composite minimality, implies local tomography.

Theorem A.3 (Local tomography, detailed version)**.** *Let V be a finite-dimensional simple Euclidean Jordan algebra equipped with the self-modeling sequential product* (12)*. Let V_{BM} be the minimal composite (Section 3.4) of two copies $V_B \cong V_M \cong V$ with the product-form sequential product. Then*

$$\dim(V_{BM}) = \dim(V_B) \cdot \dim(V_M).$$

Proof. **Step 1: Lower bound.** The product effects $\{a_i \otimes b_j\}$, where $\{a_i\}$ and $\{b_j\}$ are bases for V_B and V_M respectively, belong to V_{BM} by construction (C1). If these are linearly independent in V_{BM}, then $\dim(V_{BM}) \geq \dim(V_B) \cdot \dim(V_M)$.

Step 2: Non-degeneracy of the correlation form. Define the correlation bilinear form

$$B(a, b) = \tau\big(a \circ \varphi^{-1}(b)\big)\,,$$

where τ is the normalized trace functional on the EJA V (Faraut–Korányi [FK94], Chapter III), $\circ$ denotes the Jordan product, and $\varphi^{-1} : V_M \to V_B$ is the inverse of the tracking isomorphism.

On a simple EJA, the trace form $(a, c) \mapsto \tau(a \circ c)$ is non-degenerate (Faraut–Korányi [FK94], Proposition III.4.2): if $\tau(a \circ c) = 0$ for all $a \in V$, then $c = 0$. Since $\varphi : V_B \to V_M$ is an order isomorphism (Definition 2.6(ii)), its inverse φ^{-1} is a bijection. Therefore, if $B(a, b) = 0$ for all $a \in V_B$, then $\tau(a \circ \varphi^{-1}(b)) = 0$ for all a, which gives $\varphi^{-1}(b) = 0$ by non-degeneracy, hence $b = 0$. So B is non-degenerate.

Step 3: Linear independence of product effects. Suppose $\sum_{i,j} \alpha_{ij}\, a_i \otimes b_j = 0$ in V_{BM}, where $\{a_i\}$ and $\{b_j\}$ are bases. For any state ω on V_{BM},

evaluating on this zero element gives $\sum_{i,j} \alpha_{ij}\, \omega(a_i \otimes b_j) = 0$. In particular, for product states $\omega = \omega_B \otimes \omega_M$:

$$\sum_{i,j} \alpha_{ij}\, \omega_B(a_i)\, \omega_M(b_j) = 0 \quad \text{for all } \omega_B, \omega_M.$$

The non-degeneracy of B (which is defined via the Jordan product and trace, both available on the EJA) ensures that the matrix (α_{ij}) must be zero: fix j and vary ω_B to obtain $\sum_i \alpha_{ij}\, \omega_B(a_i) = 0$ for all ω_B; since states separate points in an OUS (Alfsen–Shultz [AS03], Theorem 1.23), this gives $\sum_i \alpha_{ij}\, a_i = 0$ for each j, hence $\alpha_{ij} = 0$ (since $\{a_i\}$ is a basis).

This establishes the lower bound: $\dim(V_{BM}) \geq \dim(V_B) \cdot \dim(V_M)$.

Step 4: Upper bound via minimality. The product effects $\{a_i \otimes b_j\}$ span a $(\dim V_B \cdot \dim V_M)$-dimensional subspace of V_{BM}. The product-form sequential product (15) preserves this subspace: $(a \otimes b) \cdot (c \otimes d) = (a \cdot c) \otimes (b \cdot d)$, which is again a product effect. The non-signaling constraints (C4) are satisfied by the product-effect subspace. By minimality (Section 3.4), V_{BM} contains no structure beyond what is forced by (C1)–(C4) and the product-form SP. Therefore $\dim(V_{BM}) \leq \dim(V_B) \cdot \dim(V_M)$.

Combining the lower and upper bounds: $\dim(V_{BM}) = \dim(V_B) \cdot \dim(V_M)$. □

Remark A.4 (Role of simplicity). Simplicity (Definition 2.6(iv)) is not needed for trace-form non-degeneracy per se: a direct sum of simple EJAs has a block-diagonal trace form that is non-degenerate on each block. Rather, simplicity ensures that the self-modeler models the *whole* system as an irreducible unit. A direct sum $V = V_1 \oplus V_2$ decomposes into superselection sectors that are invisible to each other; a system in one sector cannot model another. The theorem applies to each simple summand independently.

B Symbolic Verification

All results in this paper have been verified by an automated symbolic test suite implemented in SymPy 1.14 [MSP+17]. The test suite uses exact rational arithmetic (no floating-point approximations) and is available as supplementary material (`code/sp_verification.py`).

B.1 Test Suite Overview

The verification was conducted in two phases, corresponding to the two main stages of the derivation.

Phase 1: Sequential product axioms (S1–S7). Working on $M_2(\mathbb{C})^{\mathrm{sa}}$ (the qubit system, identified with the spin factor V_3), we verified that the corrected sequential product (12) satisfies all seven axioms of van de Wetering [vdW19b]. The test suite includes positive controls (the Lüders product $a \cdot b = \sqrt{a}\, b \sqrt{a}$ passes all axioms) and negative controls (matrix multiplication fails (S4)). A total of **186 tests** were executed:

Category	Description	Tests
S1 (additivity)	Linearity in second argument	20
S2 (continuity)	Parametric path, 11-point sweep	11
S3 (unitality)	$\mathbb{1} \cdot a = a$ for 14 effects	14
S4 (orth. symmetry)	Sharp, general, rotated effects	186*
S5 (comp. assoc.)	Compatible triples	24
S6 (comp. add.)	Complement + sum tests	8
S7 (comp. mult.)	Functional calculus	6
Non-associativity	Explicit witness + 20 random triples	22
Classical limit	Pointwise product on simplices	25
Lüders comparison	Corrected = Lüders on $M_2(\mathbb{C})^{\mathrm{sa}}$	6
φ-dependence	S4 under coarse-graining/trivial φ	18
Effect range	$0 \leq a \cdot b \leq \mathbb{1}$	49

*The S4 test count (186) includes sharp orthogonal projections (10), general effects (12), general-position rotated Bloch vectors (54), full-rank effects (4), parametric search (52), scaled projections (30), multiple φ choices (13), trivial φ (5), and Lüders positive control (6). Some tests appear in multiple categories.

Phase 2: Composite system and type exclusion. Working on $V_3 \otimes V_3$ (the body–model composite for qubits), we verified the product-form sequential product, local tomography, and the type exclusion argument. A total of **658+ tests** were executed:

Category	Description	Tests
Composite S1–S7	Axiom inheritance on $V_3 \otimes V_3$	22
Composite basics	Product-form SP, unitality, orthogonality	11
Dimension counting	$4 \times 4 = 16 = \dim(M_4(\mathbb{C})^{\mathrm{sa}})$	3
Negative checks	$\mathbb{R}$: $9 \neq 10$; $\mathbb{H}$: $36 \neq 28$	2
Classical limit	$25 \times 25 = 625$ product pairs on simplex	625
Involution P1–P4	$(X^*)^* = X$, $(XY)^* = Y^*X^*$, etc. on $M_2(\mathbb{C})^{\mathrm{sa}}$	4
Dim. formulas	$\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ at $n = 2, 3, 4$	9

B.2 Summary

In total, **844+ automated symbolic tests** were executed across both phases. All tests pass. No regressions were observed across the sequential execution of the five-plan Phase 4 and two-plan Phase 5 pipelines.

The test suite is designed as a regression harness: any modification to the sequential product formula, the composite construction, or the type exclusion arguments can be immediately checked against the full battery of tests. The use of exact SymPy arithmetic (rational numbers and algebraic expressions involving $\sqrt{3}$, $\sqrt{2}$, etc.) ensures that all verifications are exact, not approximate.

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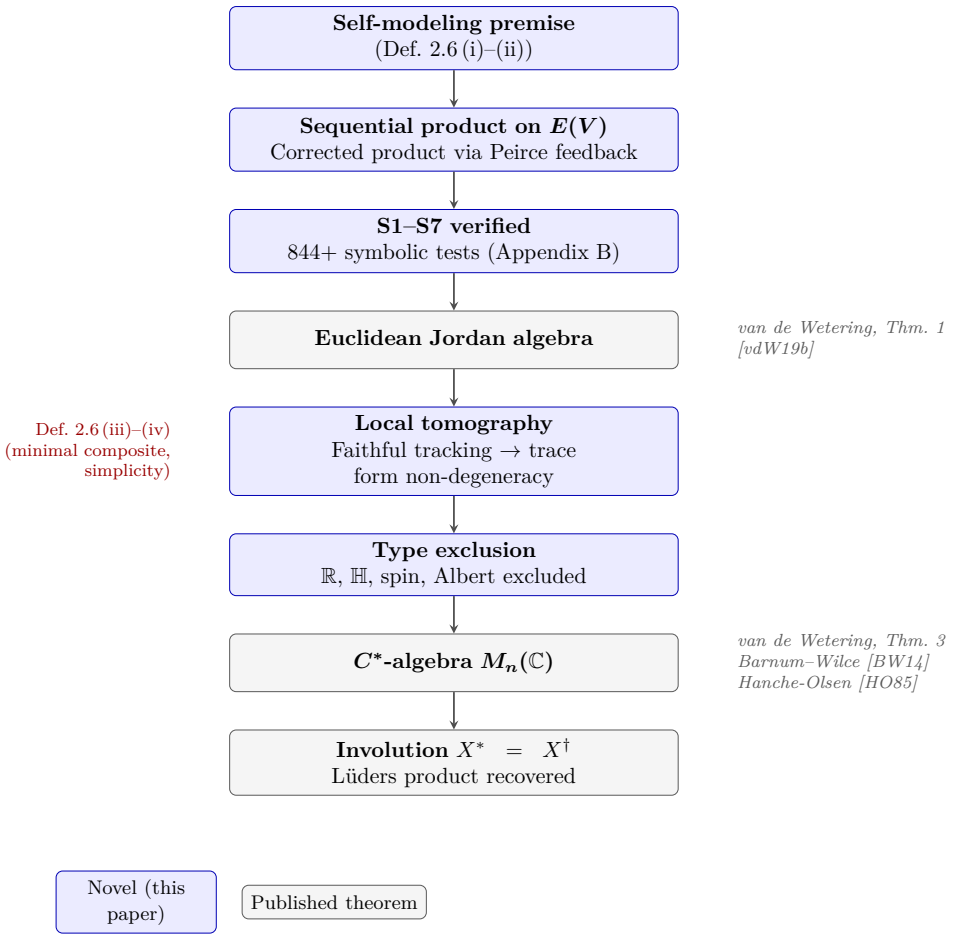

Figure 1: Logical chain of the derivation. Blue nodes indicate novel contributions of this paper; gray nodes indicate conclusions drawn from published theorems (cited). The four clauses of Definition 2.6 enter at the points indicated. Every step in the chain is either proved in the main text or deferred to Appendix A.